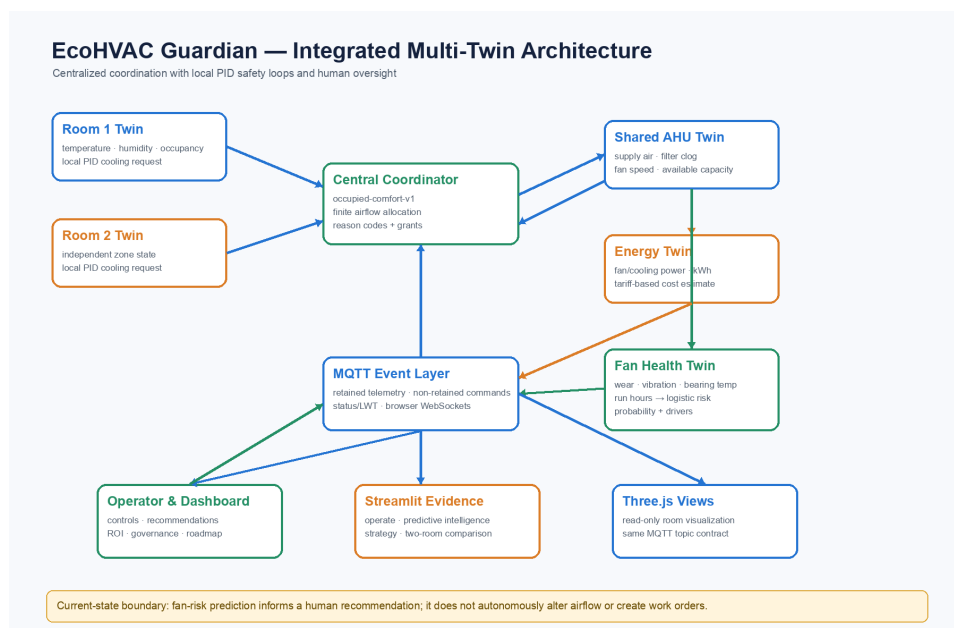


EcoHVAC Guardian: An Intelligent Two-Room Digital-Twin Ecosystem

Project 2 — Intelligent Ecosystem & Strategic Optimization



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Abstract

EcoHVAC Guardian extends a single-room teaching twin into a centrally coordinated ecosystem in which two independently controlled room twins contend for one finite-capacity air-handling unit (AHU). Each room forms an airflow request through a local PID loop. A transparent lexicographic coordinator assigns the shared supply according to occupancy and positive temperature error; the grant then updates room temperature and humidity, AHU energy, filter loading, fan condition, and an interpretable logistic failure-risk estimate. The implemented ecosystem therefore closes a deterministic sense–decide–coordinate–act–verify loop. Predictive intelligence is integrated at runtime but remains advisory: it neither changes airflow nor creates a work order.

This report derives the implemented thermal, moisture, control, allocation, degradation, energy, data-generation, training, and inference equations; records all principal units and bounds; and traces the HW2 requirements to source and evidence. At a classification threshold of 0.5, the checked-in synthetic holdout records accuracy 0.8042, balanced accuracy 0.7491, precision 0.7480, recall 0.5935, specificity 0.9046, F1 0.6619, ROC-AUC 0.8449, Brier score 0.1467, and log loss 0.4471 (92 true positives, 31 false positives, 63 false negatives, and 294 true negatives). These results show recovery of a seeded synthetic generator, not field validity: independent snapshots are labelled through a logistic relationship using the same predictors learned by the model, and the nominal seven-day target is not generated by seven-day trajectories. Live evidence demonstrates two-room resource contention and current governance/strategy interfaces. Security and ROI are explicitly bounded as, respectively, prototype controls plus a target architecture, and illustrative assumptions rather than measured outcomes.

The principal contribution is a working, explainable two-room multi-twin simulation with autonomous deterministic coordination and predictive-maintenance decision support. Five rooms are not claimed as implemented; they are analysed only as a future fairness and scalability test.

Keywords: digital twin; multi-zone HVAC; centralized coordination; PID; MQTT; logistic regression; predictive maintenance; governance; cybersecurity; ROI.

Evidence Integrity and Claim Boundary

All room physics, occupancy, equipment condition, energy, model-training data, and business inputs in this prototype are simulated or illustrative. The report does not claim real-facility calibration, seven-day field prediction, autonomous predictive maintenance, production cybersecurity, measured savings, or a five-room implementation. Bracketed author/course metadata is intentionally left for the team to complete rather than being invented. The executed notebook is present and inspected; this report does not replace its required clean rerun/provenance evidence or the unfinished Project 2 executive-pitch PPTX/PDF.

Notation and acronyms

Symbol/acronym	Meaning
AHU	Shared air-handling unit.
COP	Coefficient of performance; modeled as 3.2.
MQTT	Publish/subscribe messaging protocol used by the prototype.
PID	Proportional–integral–derivative local room controller.
$T_{i,k}$, $H_{i,k}$	Temperature (°C) and relative-humidity display value (percentage points) of room i at discrete step k .
$n_{i,k}$	Aggregate occupancy count (people).
$r_{i,k}$, $g_{i,k}$	Requested and granted supply airflow (m ³ /s).
c_k , w_k , s_k	Filter-clog, fan-wear, and normalized fan-speed fractions in $[0, 1]$ (clog is capped at 0.95).
Δt	Simulated step duration (s).
$\text{clip}(x; a, b)$	$\max(a, \min(b, x))$.
$\sigma(z)$	Logistic sigmoid, $(1 + e^{-z})^{-1}$.

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Chapter 1

Introduction

1.1 Problem and motivation

A conventional room dashboard can show a temperature and operate one controller without representing cross-zone consequences. The Project 2 problem is different: independently controlled zones share infrastructure, so one room’s request can reduce the service available to another; equipment condition changes shared capacity; energy and maintenance implications emerge from the combined state. A digital-twin ecosystem should therefore expose the referents, synchronized state, data paths, allocation authority, and physical feedback rather than present several independent cards. This interpretation follows the literature’s emphasis on connected virtual representations and bidirectional or synchronized information flows, while recognizing that the term “digital twin” is used inconsistently [5, 7].

The implemented use case is a two-room smart-laboratory simulation served by one AHU. Two rooms are the minimum non-trivial topology for demonstrating contention. The maximum room request is $0.16 \text{ m}^3/\text{s}$, whereas the nameplate AHU flow is $0.24 \text{ m}^3/\text{s}$ and the initial degraded-availability formula yields only $0.231 \text{ m}^3/\text{s}$. Thus simultaneous maximum demand, $0.32 \text{ m}^3/\text{s}$, cannot be met.

1.2 Research and engineering questions

The report addresses four linked questions:

1. Can two independently controlled room twins exhibit measurable causal interaction through a finite shared asset?
2. Can a centralized coordinator allocate scarce airflow deterministically, within physical constraints, and provide operator-readable reasons?
3. Can a trained, versioned, inspectable model be evaluated and used during each simulation tick without overstating synthetic predictive validity?
4. Can technical behavior be connected transparently to cybersecurity, ethics, ROI, and a gated deployment roadmap?

1.3 Contributions

The as-built system contributes: (i) two room state/control twins with local PID demand; (ii) a shared-AHU state model that couples comfort, energy, filter loading, and fan condition; (iii) the pure deterministic `occupied-comfort-debt-v2` allocator with bounded service debt and PID actuator feedback; (iv) a seeded NumPy logistic-regression workflow and runtime log-odds contributions; (v) retained MQTT evidence and Streamlit/Three.js views; and (vi) an explicit distinction between implemented controls and production targets.

1.4 Scope and exclusions

The report describes source state audited on 14–15 August 2026. The topology is exactly `room1` and `room2`; it is not presented as configurable N -room code. The active ecosystem uses supply-air cooling. The older direct-AC function retained in `physics.py` is documented only as a legacy compatibility path and is not substituted into the ecosystem derivation. The model is binary logistic regression, not an LLM or learned control agent. Its recommendation remains human-facing. The executed HW2 training notebook is present and inspected; a clean environment rerun remains a submission check. The report does not substitute for the still-missing final Project 2 executive-pitch PPTX/PDF; both statuses are traced explicitly.

Chapter 2

Requirements, System Boundary, and Architecture

2.1 Operational definitions

Here, an *asset/zone twin* has a referent, a state vector, update rules, interfaces, and lifecycle. Room 1, Room 2, and the shared AHU/fan satisfy this prototype definition. Energy and risk are analytical models/views derived from asset state; the coordinator is a decision service; MQTT is communication infrastructure; the dashboard and 3D clients are presentation/operator applications. This taxonomy avoids inflating every software component into a twin.

Autonomous means the simulator can repeatedly sense state, calculate room requests, allocate shared capacity, apply the grant to physical state, and publish the result without a human selecting each allocation. It does not mean that the ML model autonomously operates or maintains the AHU. *Intelligent* refers to runtime predictive inference plus context-aware, explainable coordination, not to general agency.

2.2 Requirements and acceptance criteria

Table 2.1: HW2 requirement interpretation and auditable acceptance criteria.

ID	Brief requirement	Acceptance criterion	As-built evidence/status
R1	Intelligent/autonomous ecosystem	Repeating closed state-to-decision-to-physical-state loop	Local PID, coordinator, supply-air physics; met for deterministic control.
R2	Failure-prediction model	Versioned trained model produces runtime risk	JSON logistic artifact scored each tick; met with synthetic validity boundary.
R3	Multiple interacting twins	One twin's demand changes another's grant through a shared constraint	Two-room stress evidence; met.
R4	Data-flow map and coordination strategy	Diagram names flows and central/federated strategy	Integrated architecture and centralized allocator; met.

ID	Brief requirement	Acceptance criterion	As-built evidence/status
R5	Cybersecurity measures	Current controls and target controls are separated and threat-led	Strict validation, retained-command rejection, correlated results, bounded replay cache, optional local hash-chain command audit, and LWT are current; TLS/identity/ACL files are undeployed examples; partial .
R6	ROI analysis	Equations, inputs, outputs, provenance, sensitivity	Calculator and illustrative scenarios; no measured baseline; partial .
R7	Transparency and bias mitigation	Model/policy explanation plus operational mitigation process	Coefficients, contributions, reason codes; subgroup thresholds/process planned; partial .
R8	Pilot-to-scale roadmap	Stages, owners, gates, KPIs, rollback	Five-stage governed plan; documented, not executed .
R9	Notebook or AutoML result	Executable named deliverable included	Fully executed <code>notebooks/fan_failure_prediction.ipynb</code> is present and inspected; clean submission-environment rerun remains; met as present .
R10	Executive pitch	Current Project 2 PPTX plus exported PDF focused on value, evidence, security, ethics, and the pilot ask	Markdown outline only; final Project 2 deck/export absent; Project 1 PPTX/video are legacy and invalid as Project 2 evidence; not met .

2.3 Submission deliverables register

This report does not substitute for verifying the executed HW2 training notebook in the submission environment or for producing the final executive-pitch PPTX/PDF. The register separates files that exist from work still needed; “current” means found in the audited project/package, not independently approved by the instructor.

Table 2.2: Named submission assets and closure actions.

Deliverable	Current status	Action and acceptance evidence
Academic report PDF and portable source	Current: PDF, TeX, bibliography, figures, build files, source ZIP, and hashes	Complete metadata; clean-build from extracted ZIP; submit current PDF/source with verified manifest.
Executed model notebook or accepted AutoML output	Current: 23-cell notebook with all 11 code cells executed; includes EDA, baseline/expanded metrics, confusion matrix, calibration bins, OOD/abstention examples, limitations, and exact artifact checks	Verify a clean rerun in the submitted environment; preserve outputs, environment/provenance, and byte-for-byte artifact/metric checks.

Deliverable	Current status	Action and acceptance evidence
Ecosystem/data-flow diagram	Current: integrated two-room architecture figure	Verify legibility and consistency with exactly two rooms, centralized allocation, active supply-air model, and advisory ML boundary.
Project 2 executive pitch PPTX and exported PDF	Current: editable generator, eight-slide PPTX, eight-page PDF, evidence index, and rendered verification under <code>report/pitch/</code>	Preserve source-relative assets and final hashes; team must still name the pitch owner and rehearse against the assigned time. Legacy Project 1 PPTX and videos remain invalid evidence.
Current evidence screenshots	Current: five Project 2 figures copied into the source package	Retain provenance; ensure captions distinguish state evidence from intervention/repeated-run proof.
Model artifacts	Current: versioned JSON coefficients and stored metrics; synthetic only	Hash and include artifact, metrics, generator/trainer version, and executed notebook evidence; do not present audit-derived metrics as stored.
Test evidence	Current: recorded run: 145 tests pass; raw console/JUnit, environment, hashes, and real-broker stress/pause/baseline evidence are archived under <code>release-evidence/project2/</code>	Refresh source-state and hashes after the final commit; add coverage, browser automation, hardened-client integration, and reviewer disposition.
Reproducibility manifest	Current: build commands, source ZIP, and SHA256SUMS	Verify hashes after final metadata/content changes and prove a clean build from the extracted archive without parent files.

2.4 Logical architecture

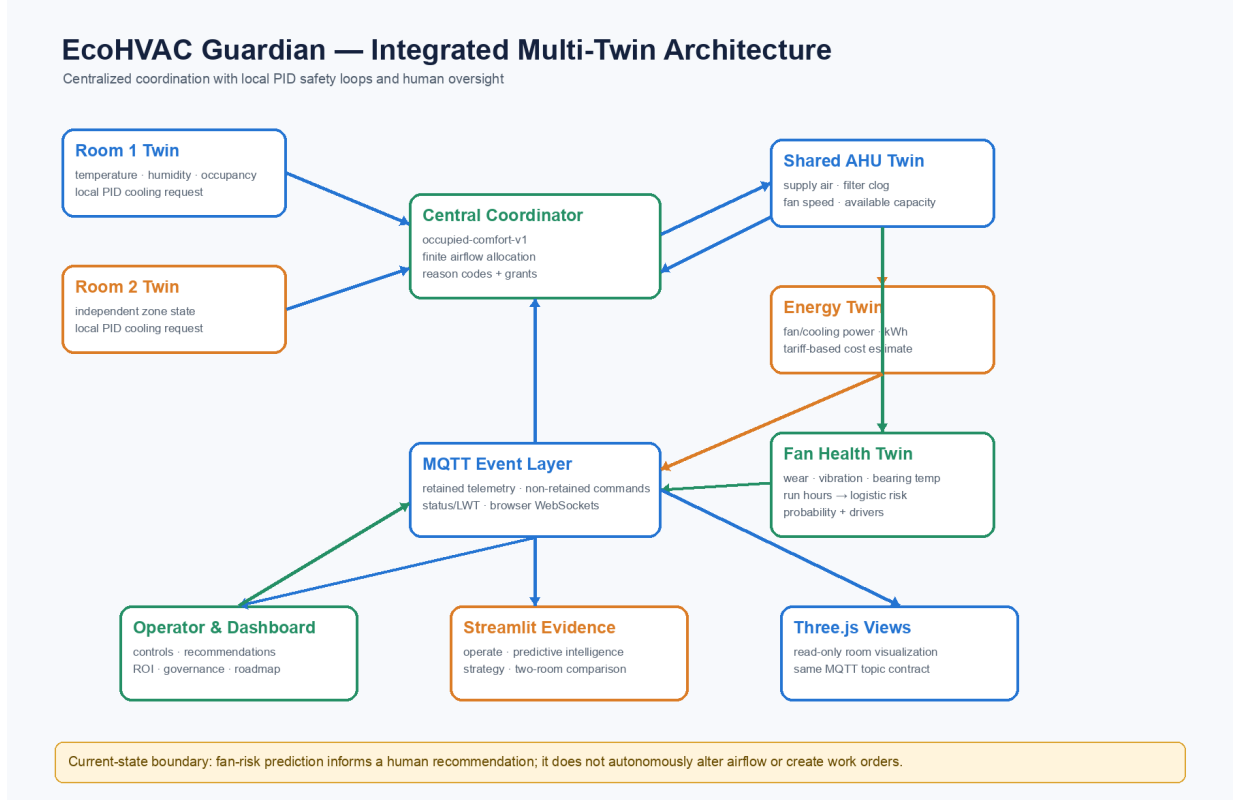


Figure 2.1: As-built two-room logical architecture. Green/blue data paths link local control, centralized allocation, shared-AHU physics, fan risk, MQTT, and evidence clients. Prediction is intentionally marked advisory.

The coordinator is centralized because one service has the complete demand set and controls one indivisible shared capacity. PID remains local to preserve per-zone state and response. A federated strategy is not implemented and would be relevant only across multiple AHUs/buildings or when local operation must survive loss of a higher-level coordinator.

2.5 State, interfaces, and update order

For room i , a useful abstract state is

$$\mathbf{x}_{i,k} = [T_{i,k}, H_{i,k}, n_{i,k}, T_{i,k}^{sp}, h_{i,k}, u_{i,k}, r_{i,k}, g_{i,k}]^T. \quad (2.1)$$

The AHU state includes maximum and available airflow, supply temperature, clog, wear, speed, instantaneous power, and cumulative energy. Fan-health state includes run hours, wear, vibration, bearing temperature, and health.

One `tick` executes in this exact order: (1) auto-HVAC and PID requests; (2) available-capacity calculation; (3) centralized allocation; (4) room temperature/humidity updates using granted flow; (5) AHU filter/energy update; (6) fan-state update; (7) logistic-risk inference; (8) coherent snapshot return and periodic MQTT publication. Risk therefore follows the current allocation and cannot affect it.

2.6 Communication and deployment boundary

The Paho client publishes retained state and telemetry so late subscribers obtain the latest evidence. Retained commands are rejected before queueing. Non-retained commands are queued in the MQTT callback before mutation on the simulator thread; correlated application results distinguish accepted, changed, replayed, and conflicting command IDs. MQTT semantics and retained/LWT behavior are protocol features, not security controls [10]. Current Compose packages only Mosquitto; simulator, dashboard, and static 3D server remain host processes. The publisher and browser use `localhost`; this is suitable for a local demonstration, not remote deployment.

Chapter 3

Physical, Control, Degradation, and Energy Models

3.1 Discrete-time assumptions

All models are lightweight teaching proxies, not calibrated building models. The simulator uses a non-negative step Δt in seconds, normally selected from $\{1, 2, 5, 10\}$. State clipping prevents numerical values outside demonstration bounds. Occupancy may advance through a bounded stochastic random walk after room physics; deterministic scenario analysis should disable or seed it. Reported values are rounded only at MQTT/display boundaries.

3.2 Local PID request

The temperature error is positive when a room is hot:

$$e_{i,k} = T_{i,k} - T_{i,k}^{sp}. \quad (3.1)$$

The implemented discrete controller is

$$I_{i,k} = \text{clip}(I_{i,k-1} + e_{i,k}\Delta t; -20, 20), \quad (3.2)$$

$$D_{i,k} = \begin{cases} (e_{i,k} - e_{i,k-1})/\Delta t, & \Delta t > 0, \\ 0, & \Delta t = 0, \end{cases} \quad (3.3)$$

$$u_{i,k} = \text{clip}(0.40e_{i,k} + 0.05I_{i,k} + 0.05D_{i,k}; 0, 1), \quad (3.4)$$

$$r_{i,k} = 0.16h_{i,k}u_{i,k} \quad [\text{m}^3/\text{s}]. \quad (3.5)$$

On first use, previous error is initialized to current error, so the derivative is zero. Integral clamping bounds its contribution to ± 1 . After allocation, `apply_actuator_feedback` converts granted flow to applied output $u_i^{app} = g_i/0.16$ and bleeds the integral when $u_i > u_i^{app}$:

$$I_i \leftarrow \text{clip}\left(I_i - \frac{(u_i - u_i^{app})\Delta t}{10K_i}; -20, 20\right), \quad K_i > 0. \quad (3.6)$$

This back-calculation-style feedback reduces windup under shared-actuator saturation; it does not itself guarantee fairness or eliminate a sustained full request under persistent heat load. PID fundamentals and anti-windup concerns are discussed in Åström and Hägglund [1].

Auto mode switches cooling off at zero occupancy and on when an occupied room is above setpoint; otherwise it keeps the previous binary state. Setpoints are clamped to 18–30 °C.

3.3 Available shared airflow

With nameplate flow $F_{\max} = 0.24 \text{ m}^3/\text{s}$, clog fraction $c_k \in [0, 0.95]$, and wear fraction $w_k \in [0, 1]$,

$$F_k^{\text{avail}} = F_{\max}(1 - 0.65c_k)(1 - 0.15w_k). \quad (3.7)$$

At defaults $c_0 = 0.05$, $w_0 = 0.03$, [equation \(3.7\)](#) gives $0.231\,155\,1 \text{ m}^3/\text{s}$; at maximum modeled degradation it gives $0.078\,03 \text{ m}^3/\text{s}$. Names ending in `_pct` for clog, wear, and speed are fractions, not numbers in $[0, 100]$; the dashboard formats them as percentages. Humidity and health, in contrast, are percentage-point quantities.

3.4 Active supply-air thermal equation

The active ecosystem computes occupant and envelope loads and sensible supply-air cooling:

$$Q_{i,k}^{\text{people}} = n_{i,k}q_p, \quad q_p = 100 \text{ W/person}, \quad (3.8)$$

$$Q_{i,k}^{\text{wall}} = K_w(T_o - T_{i,k}), \quad K_w = 0.05 \text{ W/}^\circ\text{C}, \quad T_o = 32^\circ\text{C}, \quad (3.9)$$

$$Q_{i,k}^{\text{cool}} = \rho_a c_{p,a} g_{i,k} \max(0, T_{i,k} - T_s), \quad (3.10)$$

where $\rho_a = 1.2 \text{ kg/m}^3$, $c_{p,a} = 1005 \text{ J/(kg }^\circ\text{C)}$, and $T_s = 16^\circ\text{C}$. Thus $\rho_a c_{p,a} = 1206 \text{ J/(m}^3 \text{ }^\circ\text{C)}$. With lumped heat capacity $C_r = 25\,000 \text{ J/}^\circ\text{C}$,

$$T_{i,k+1} = \text{clip}\left(T_{i,k} + \frac{\Delta t}{C_r}[Q_{i,k}^{\text{people}} + Q_{i,k}^{\text{wall}} - Q_{i,k}^{\text{cool}}]; 15, 40\right). \quad (3.11)$$

The very low nominal envelope conductance and heat capacity are classroom assumptions. No infiltration, solar gain, thermal mass separation, duct losses, latent psychrometrics, or sensor dynamics are calibrated. Accordingly, the output demonstrates causal logic, not compliance with a comfort standard such as ASHRAE 55 [\[2\]](#).

Legacy distinction. `physics.py` retains an older direct-AC equation with a $-3500h_i u_i \text{ W}$ term. `EcosystemSimulator.tick` calls `step_temperature_from_supply_air`, so [equation \(3.11\)](#)—not the legacy direct term—is the report’s active model.

3.5 Humidity proxy

$$H_{i,k+1} = \text{clip}(H_{i,k} + \Delta t(0.06n_{i,k} - 2.1g_{i,k}); 15, 80). \quad (3.12)$$

Here $0.06n$ is an empirical occupancy moisture increment and $2.1g$ an airflow-driven drying rate in units required to produce percentage points per second. This is not a humidity-ratio or latent-load calculation and should not be interpreted psychrometrically.

3.6 Filter, fan condition, and energy

Total delivered flow and normalized speed are

$$G_k = \sum_i g_{i,k}, \quad s_k = \text{clip}(G_k/F_{\max}; 0, 1). \quad (3.13)$$

The filter grows as

$$c_{k+1} = \text{clip}(c_k + 2.5 \times 10^{-6} s_k \max(0, \Delta t); 0, 0.95). \quad (3.14)$$

Fan wear and derived telemetry are

$$w_{k+1} = \text{clip}[w_k + \Delta t s_k (8 \times 10^{-7} + 1.5 \times 10^{-6} c_{k+1}); 0, 1], \quad (3.15)$$

$$T_{b,k+1}^* = 31 + 23s_k + 13c_{k+1} + 12w_{k+1}, \quad (3.16)$$

$$T_{b,k+1} = T_{b,k} + \min(1, \Delta t/45)(T_{b,k+1}^* - T_{b,k}), \quad (3.17)$$

$$v_{k+1} = 0.8 + 4w_{k+1} + 1.4c_{k+1}s_k, \quad (3.18)$$

$$h_{k+1}^{fan} = \text{clip}(100 - 72w_{k+1} - 20c_{k+1}; 0, 100), \quad (3.19)$$

$$t_{k+1}^{run} = t_k^{run} + \Delta t s_k / 3600. \quad (3.20)$$

These are empirical simulation relations, not identified degradation laws. The c_{k+1} indices are intentional: **advance_ahu** updates filter loading before **step_fan_state** receives the clog value. Fan speed s_k and energy, in contrast, are calculated from the just-completed allocation using pre-update AHU condition.

The modeled fan and cooling electrical powers are

$$P_k^{fan} = 180s_k^3(1 + 1.8c_k)(1 + 0.25w_k), \quad (3.21)$$

$$P_k^{cool,e} = \frac{\sum_i Q_{i,k}^{cool}}{3.2}, \quad (3.22)$$

$$P_k^{total} = P_k^{fan} + P_k^{cool,e}, \quad (3.23)$$

$$E_{k+1} = E_k + \frac{P_k^{total} \Delta t}{3.6 \times 10^6} \quad [\text{kW h}]. \quad (3.24)$$

The cubic relation reflects the familiar fan-system affinity-law trend, while actual systems depend on their operating curve and controls [14]. Importantly, clog does *not* unconditionally increase modeled total power. At fixed s , the resistance multiplier in equation (3.21) increases with c ; under saturated demand, however, clog reduces available flow and therefore s^3 , which can dominate. The report therefore claims only increased modeled resistance at fixed flow.

3.7 Timing consistency

The room runtime stores the exact pre-state sensible thermal cooling used by the room update and AHU calculation, and MQTT publishes that stored value as **thermal_cooling_power_w**. AHU telemetry separately publishes fan electrical power, COP-derived cooling electrical power, total electrical power, and cumulative electrical energy. Thus the earlier pre/post-step recomputation discrepancy is removed and thermal versus electrical quantities are explicit; they remain uncalibrated simulation estimates rather than facility measurements.

Chapter 4

Centralized Coordination

4.1 Priority and allocation

For enabled room i , positive comfort error and request are

$$e_i^+ = \max(0, T_i - T_i^{sp}), \quad r_i = \max(0, r_i^{raw}). \quad (4.1)$$

The implemented `occupied-comfort-debt-v2` priority tuple is

$$\pi_i = (\mathbb{I}[n_i > 0], e_i^+, d_i, n_i, \text{roomID}_i), \quad (4.2)$$

where d_i is bounded accumulated comfort debt in °C s. Sorting is descending on the first four fields and ascending lexical room ID. Let $(1), \dots, (N)$ be that order. Starting with $R_0 = F^{avail}$,

$$g_{(j)} = \min(r_{(j)}, R_{j-1}), \quad R_j = R_{j-1} - g_{(j)}. \quad (4.3)$$

Disabled rooms request zero. Capacity is flagged constrained when $\sum_i r_i > F^{avail} + 10^{-9}$.

4.2 Comfort-debt state and actuator feedback

For an enabled occupied room with positive request, limited service adds

$$d_{i,k+1} = \text{clip} \left[d_{i,k} + e_i^+ \left(\frac{r_i - g_i}{r_i} \right) \Delta t; 0, 3600 \right] \quad (4.4)$$

and increments consecutive limited-service time by Δt . Full service recovers debt at 2 °C s per simulated second and resets consecutive limited-service time; disabled, unoccupied, or no-request states also recover debt and reset that timer. Debt enters priority only after current occupancy and temperature error, so it mitigates near-tie starvation but cannot overrule a larger current error.

The grant is also fed back to the local PID as applied output $g_i/0.16$. This reduces integral windup when the shared actuator cannot deliver the requested output; it does not make allocation proportional or guarantee minimum service.

4.3 Safety invariants and replayability

Because each update is a minimum with a non-negative remainder, the pure function guarantees

$$0 \leq g_i \leq r_i, \quad \sum_i g_i \leq F^{avail}. \quad (4.5)$$

The same demand list, including debt state, and capacity produce the same decisions, scores, and reason codes. This makes allocation unit-testable and replayable. Sorting dominates complexity at $O(N \log N)$; allocation is $O(N)$.

4.4 Interpretation and limits

The rule can be interpreted as lexicographically maximizing the service sequence in ranked order, but it is not a scalar multi-objective optimization. It has no explicit energy, risk, minimum-ventilation, switching-cost, or guaranteed proportional-fairness term. An occupied room with a larger current error may consume all remaining flow despite another room's debt. Equal current conditions may be influenced by debt; if debt and occupancy also tie, stable room ID remains the final resolver.

Appropriate reported metrics include room allocation ratio g_i/r_i , comfort debt, consecutive limited-service time, denied-flow integral $\sum_k (r_i - g_i) \Delta t$, degree-hours outside the comfort band, and maximum room disparity. Production policy would additionally need explicit minimum ventilation and owner-approved comfort/starvation limits.

4.5 Why two rooms are sufficient

Two rooms already create the non-trivial inequality $r_1 + r_2 > F^{avail}$ and expose every core mechanism: cross-twin causality, capacity conservation, priority, a full or partial grant, a zero grant, debt accumulation/recovery, comfort consequences, and shared fan/energy state. Five copies without a new hypothesis would add UI surface, not intelligence.

For future equal maximum requests,

$$R_{max}(N) = 0.16N, \quad \phi_N = F^{avail}/(0.16N). \quad (4.6)$$

At initial availability, $\phi_2 = 0.7224$ and $\phi_5 = 0.2889$. At $c = 0.85, w = 0.75$, $F^{avail} = 0.0953175 \text{ m}^3/\text{s}$, giving $\phi_2 = 0.2979$ and $\phi_5 = 0.1191$. These are aggregate ratios; under greedy allocation some lower-ranked rooms receive zero. A five-room extension is therefore justified only as a fairness, latency, message-load, and scalability experiment accompanied by generalized configuration and tests.

Chapter 5

Predictive-Maintenance Model

5.1 Task and evidence boundary

The operational feature vector is

$$\mathbf{x} = (c, s, v, T_b, t^{run})^\top, \quad (5.1)$$

comprising filter-clog fraction, fan-speed fraction, vibration (mm/s), bearing temperature (°C), and run hours. The binary target is named `failure_within_7d`. That name is a declared synthetic maintenance horizon: no asset trajectory is advanced seven days, and no observed future event, censoring, maintenance reset, or equipment ID exists. The model estimates the synthetic label relationship, not a calibrated real-fan probability.

5.2 Seeded synthetic generator

The generator creates 2,400 independent rows with seed 20260805:

$$c \sim U(0, 0.95), \quad s \sim U(0.08, 1), \quad t^{run} \sim U(0, 9000), \quad (5.2)$$

$$\epsilon_w \sim U(-0.08, 0.08), \quad w = \min(1, 0.05 + t^{run}/15000 + \epsilon_w), \quad (5.3)$$

$$\epsilon_v \sim \mathcal{N}(0, 0.22^2), \quad v = \max(0.4, 0.7 + 3.9w + 1.35cs + \epsilon_v), \quad (5.4)$$

$$\epsilon_T \sim \mathcal{N}(0, 1.4^2), \quad T_b = 30 + 22s + 15c + 15w + \epsilon_T. \quad (5.5)$$

The hidden label relation is

$$\eta^{gen} = -6.3 + 1.6c + 0.9s + 0.75(v - 1) \quad (5.6)$$

$$+ 0.105(T_b - 35) + 0.00014t^{run}, \quad (5.7)$$

$$p^{gen} = \sigma[\text{clip}(\eta^{gen}; -50, 50)], \quad y \sim \text{Bernoulli}(p^{gen}). \quad (5.8)$$

The Bernoulli draw uses the unrounded generated values. The five predictors are then rounded to six decimal places before they are stored and passed to training. The generator’s wear expression has an upper but no explicit lower clamp; given its specified ranges, slightly negative wear is possible near zero run hours. This is a dataset-quality limitation.

5.3 Training algorithm

A seeded permutation creates 1,920 training and 480 holdout rows. Means μ_j and population standard deviations σ_j are calculated from training rows only; scales below 10^{-9} are replaced with one:

$$z_{ij} = (x_{ij} - \mu_j)/\sigma_j. \quad (5.9)$$

Starting with zero coefficients/intercept, 2,400 full-batch gradient steps at learning rate $\alpha = 0.08$ minimize unregularized binary cross-entropy

$$\mathcal{L}(\beta, b) = -\frac{1}{m} \sum_{i=1}^m [y_i \log p_i + (1 - y_i) \log(1 - p_i)], \quad (5.10)$$

using

$$\mathbf{p} = \sigma(Z\beta + b), \quad (5.11)$$

$$\beta \leftarrow \beta - 0.08 Z^\top (\mathbf{p} - \mathbf{y}) / m, \quad (5.12)$$

$$b \leftarrow b - 0.08 \text{mean}(\mathbf{p} - \mathbf{y}). \quad (5.13)$$

This is conventional binary logistic regression [3]. There is no conventional standardization leakage, because holdout rows do not define the training means/scales. There is nevertheless *generator circularity*: labels arise from these same predictors and a logistic family that the learner is designed to recover.

5.4 Stored model and exact runtime equation

Table 5.1: Checked-in model parameters. Means and scales are train-set statistics; coefficients act on standardized features.

Feature j	μ_j	σ_j	β_j
Filter clog fraction	0.48097131	0.27703595	0.52486318
Fan speed fraction	0.54311963	0.26961559	0.32268269
Vibration (mm/s)	2.39246047	0.79832257	0.72890271
Bearing temperature (°C)	54.29740422	7.94594929	0.65599373
Run hours	4392.58951949	2595.06254848	0.39002046

With intercept $b = -1.45350261$, the deployed logit is

$$\eta = -1.45350261 + 0.52486318 \frac{c - 0.48097131}{0.27703595} + 0.32268269 \frac{s - 0.54311963}{0.26961559} \quad (5.14)$$

$$+ 0.72890271 \frac{v - 2.39246047}{0.79832257} + 0.65599373 \frac{T_b - 54.29740422}{7.94594929} \quad (5.15)$$

$$+ 0.39002046 \frac{t^{run} - 4392.58951949}{2595.06254848}, \quad (5.16)$$

$$\hat{p}_{\text{fail}} = \sigma[\text{clip}(\eta; -50, 50)]. \quad (5.17)$$

The artifact version is **fan-risk-logistic-v1**. The trainer also persists per-feature domain bounds: physical limits intersected with the training minimum/maximum expanded by 30% of its range. Runtime scoring first requires all five finite numeric inputs inside those bounds; otherwise it returns no probability and an explicit abstained or out-of-distribution status. A missing/corrupt default artifact loads an unavailable non-predicting fallback. These are safe non-prediction states, not evidence of low risk. Runtime bands for an in-domain score are

$$\text{band}(p) = \begin{cases} \text{low}, & p < 0.35, \\ \text{medium}, & 0.35 \leq p < 0.65, \\ \text{high}, & p \geq 0.65. \end{cases} \quad (5.18)$$

These operating labels are distinct from the 0.5 cutoff used to calculate stored classification metrics. Conflating the two would make the dashboard threshold and reported confusion matrix appear inconsistent.

5.5 Explainability

For each feature, the displayed contribution is

$$d_j = \beta_j(x_j - \mu_j)/\sigma_j, \quad \eta = b + \sum_j d_j. \quad (5.19)$$

It is an additive log-odds decomposition, not a causal effect. Correlated generated features (wear influences vibration and temperature) mean contributions should not be interpreted independently. The code selects the three largest *signed* d_j , not the largest magnitudes; a strong negative risk-reducing contribution can therefore be omitted. A fuller UI would show the intercept, all five signed terms, and the final sigmoid.

5.6 Holdout results

Table 5.2: Holdout confusion matrix at classification cutoff 0.5 ($n = 480$).

	Predicted negative	Predicted positive
Actual negative	TN = 294	FP = 31
Actual positive	FN = 63	TP = 92

Stored metrics are

$$\text{Accuracy} = (TP + TN)/480 = 0.8042, \quad (5.20)$$

$$\text{Precision} = TP/(TP + FP) = 0.7480, \quad (5.21)$$

$$\text{Recall} = TP/(TP + FN) = 0.5935. \quad (5.22)$$

The 63 false negatives are operationally material. The current metrics artifact stores the following expanded measures (earlier versions stored only the basic counts/rates):

$$\text{Specificity} = TN/(TN + FP) = 0.9046, \quad (5.23)$$

$$F_1 = 2TP/(2TP + FP + FN) = 0.6619, \quad (5.24)$$

$$\text{NPV} = TN/(TN + FN) = 0.8235, \quad (5.25)$$

$$\text{Balanced accuracy} = (\text{Recall} + \text{Specificity})/2 = 0.7491. \quad (5.26)$$

Holdout positive prevalence is $155/480 = 0.3229$, so an always-negative classifier scores 0.6771 accuracy; model accuracy is 0.1271 higher. Accuracy alone is insufficient, and PR curves are particularly useful under unequal class prevalence [12]. The stored artifact now includes ROC-AUC 0.8449, log loss 0.4471, and Brier score 0.1467; the executed notebook adds the majority baseline, confusion matrix, and calibration bins. PR-AUC/average precision, confidence intervals, cost-based threshold selection, and field calibration remain absent. Brier score assesses probability error [4]; synthetic calibration bins do not establish calibration on real assets [6].

5.7 Validity and deployment gaps

One random holdout has no temporal or asset-group separation, repeated seeds, confidence intervals, external data, or prospective drift evaluation. Missing/non-finite inputs and univariate domain violations now cause explicit abstention; however, in-bound multivariate combinations can still be unlike training rows. Thresholds are not selected from maintenance costs. The executed notebook provides EDA, a majority baseline, expanded metrics, calibration bins, OOD examples, limitations, and exact artifact reproduction. The remaining evidence upgrade is

realistic asset/time trajectories, grouped or temporal validation, PR analysis, bootstrap intervals, threshold-cost analysis, multivariate drift/OOD behavior, and a prospectively governed model card [8].

Chapter 6

Experimental Evidence and Results

6.1 Evidence protocol

The figures were captured from the then-current Project 2 source during a local run on 14 August 2026. They are state evidence, not a substitute for a machine-readable experiment log. The stress capture was created using validated individual room/AHU MQTT commands. The current implementation now supports correlated `twin/ecosystem/cmd/scenario` baseline/stress commands and retained scenario/application-result topics; this historical image is not retroactively evidence that the newer guided button caused the captured state.

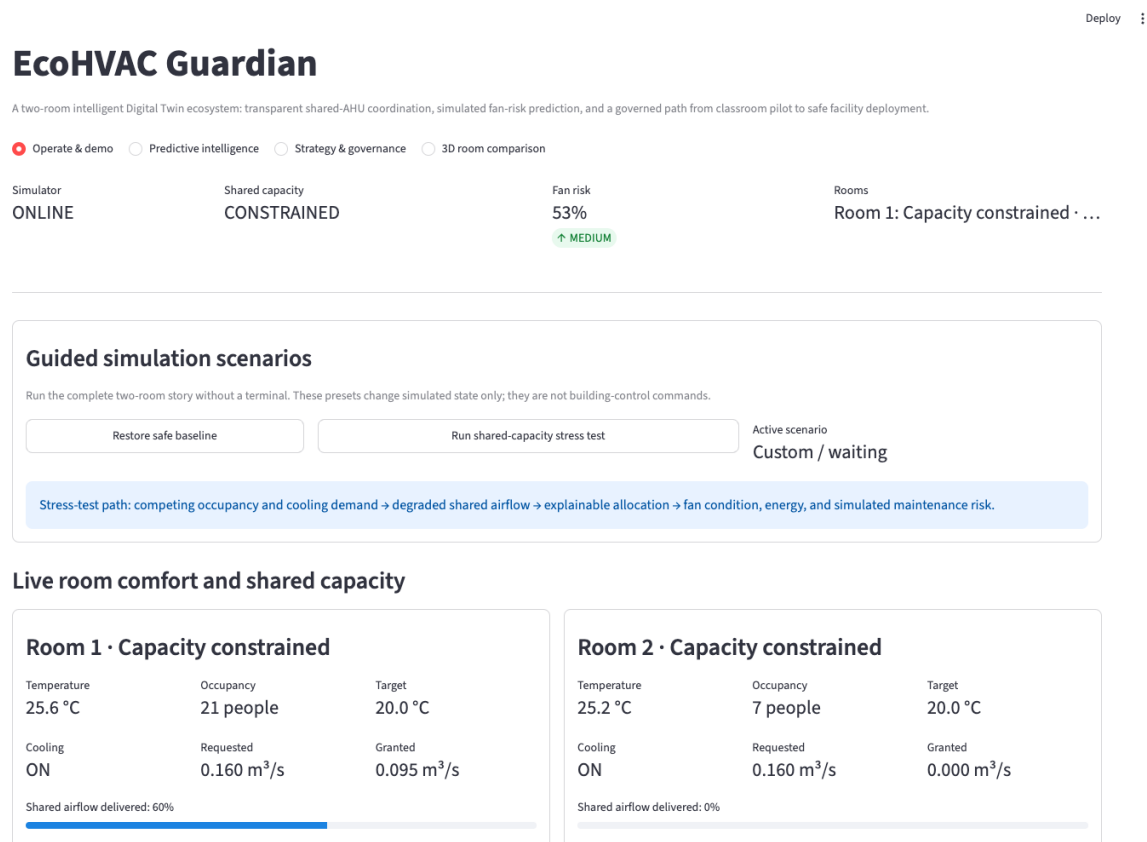
For a repeatable submission experiment, occupancy randomness should be seeded or disabled, Δt , initial state, command sequence, duration, artifact/model hashes, and raw MQTT decisions should be recorded. The expected invariants are $0 \leq g_i \leq r_i$, $\sum g_i \leq F^{avail}$, and deterministic priorities from the same snapshot.

6.2 Scenario definitions

Table 6.1: Recommended reproducible scenario protocol. E1–E3 use implemented command paths; current captures directly evidence E2/E3 state but not repeated trials.

ID	Initial/command condition	Hypothesis and measures	Acceptance condition
E1	Clean/default AHU; aggregate request no greater than available flow	All requests granted; record request, grant, temperature, power	$g_i = r_i$ and unconstrained flag
E2	Both rooms hot/occupied with aggregate request above capacity	Priority creates unequal grants while conserving capacity	deterministic ranking; grant sum within capacity
E3	Same demand, inject $c = 0.85, w = 0.75$	Capacity falls; denied flow, comfort, fan telemetry, and risk change	$F^{avail} = 0.0953175 \text{ m}^3/\text{s}$ and complete telemetry
E4	Equal demand/occupancy/error for both rooms	Stable room-ID tie is exposed	repeatable Room 1-first result; fairness limitation recorded
E5	Model loss/OOD, malformed/retained/replayed command, or simulation stop	Reject unsafe command, explicitly abstain/unavailable, and freeze instantaneous actuation on stop	unit-tested; browser plus real-broker end-to-end evidence remains

6.3 Observed shared-capacity state



6.4 Observed predictive workspace

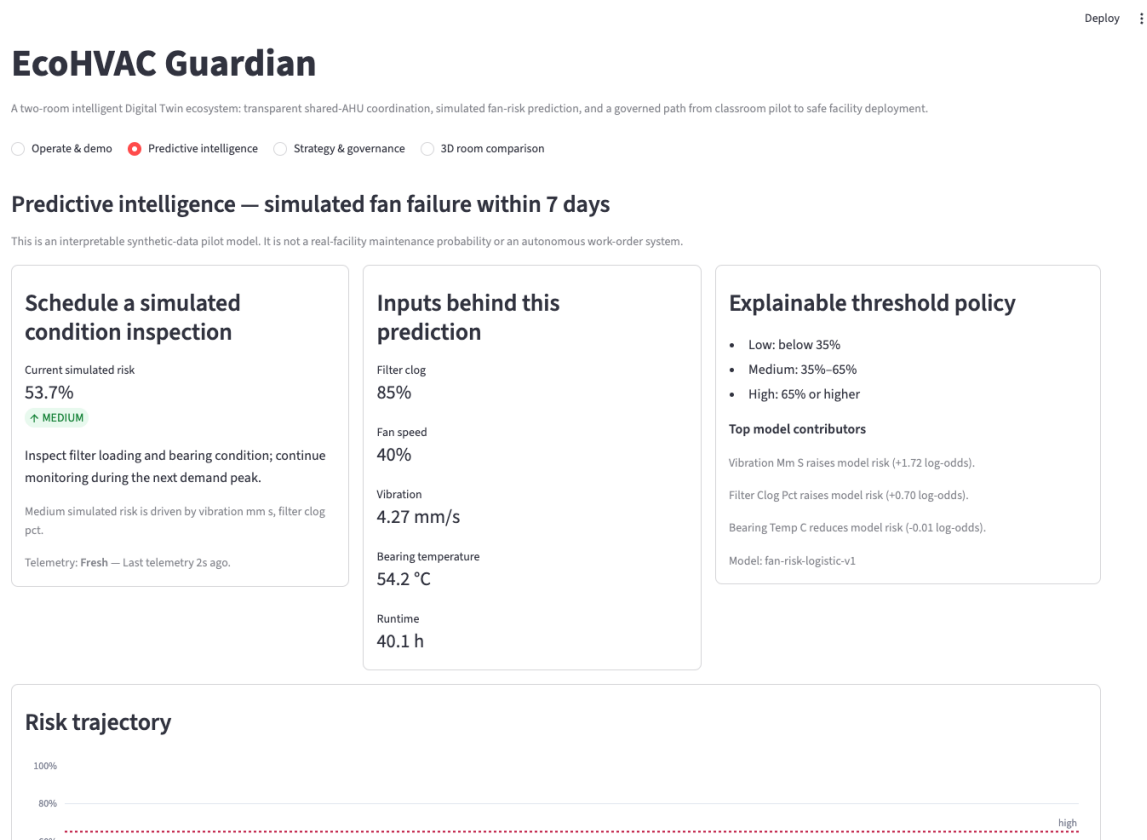


Figure 6.2: Runtime predictive-intelligence workspace showing 53.7% medium simulated risk, input telemetry, model version, signed contributors, thresholds, freshness, and a human inspection recommendation.

This provides existence and integration evidence: the loaded model consumes live fan telemetry after each tick. It does not validate the number as a real probability, nor does the recommendation become a work order or control input.

6.5 Strategy and simultaneous visualization

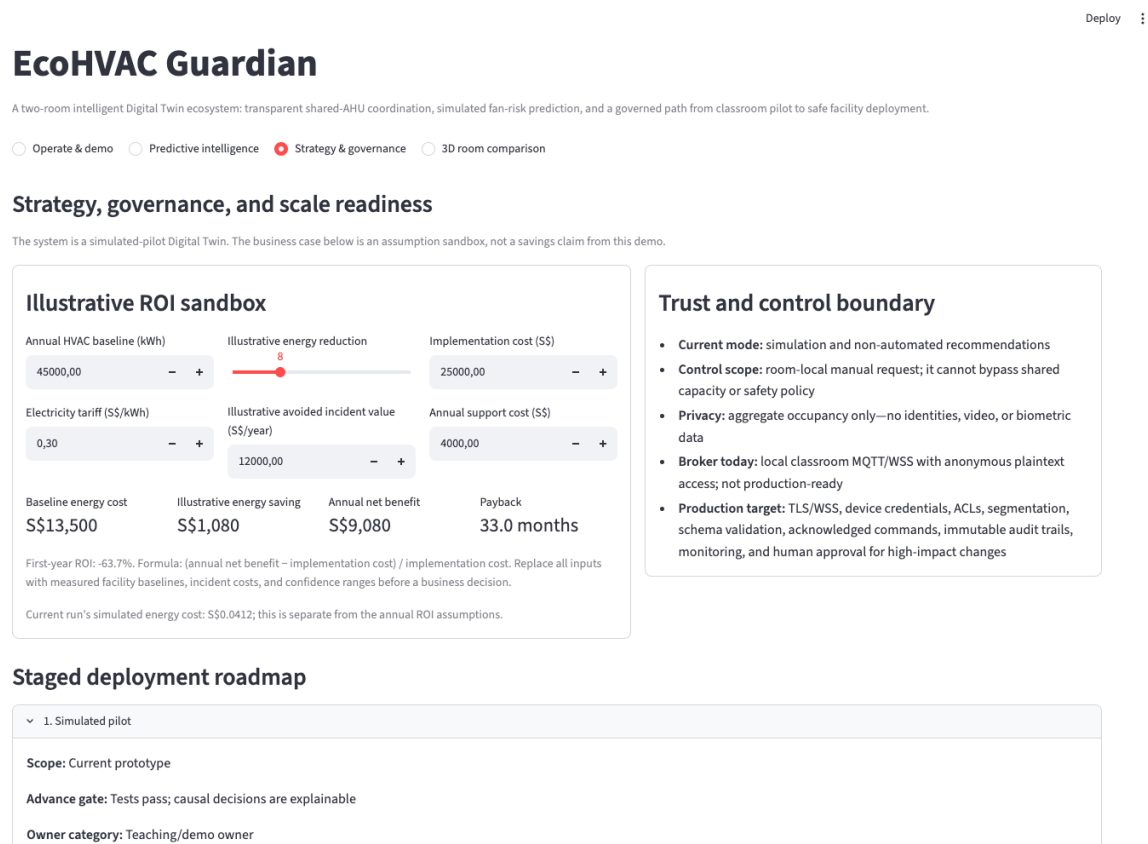


Figure 6.3: Strategy/governance workspace: editable illustrative ROI, current versus target security, and staged deployment material. Values are assumptions, not observed savings.

Deploy ⋮

EcoHVAC Guardian

A two-room intelligent Digital Twin ecosystem: transparent shared-AHU coordination, simulated fan-risk prediction, and a governed path from classroom pilot to safe facility deployment.

☐ Operate & demo
 ☐ Predictive intelligence
 ☐ Strategy & governance
 ☒ 3D room comparison

3D room comparison

Both room twins are visible simultaneously for visual comparison. These are read-only views using the same retained MQTT topic contract; use Operate & demo for commands.

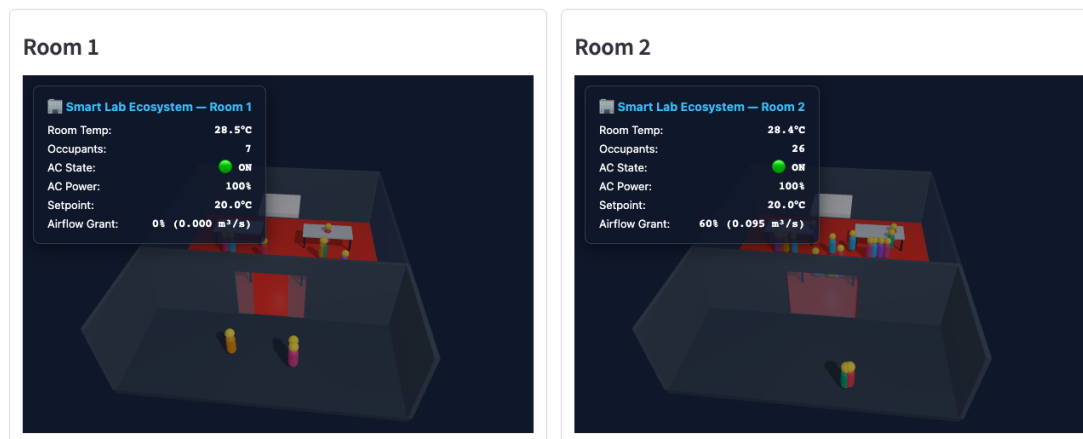


Figure 6.4: Simultaneous room-specific Three.js views consuming retained topics. Visualization is useful evidence of distinct room state but is not itself evidence of intelligent allocation.

The 3D client supports only `room1` and `room2`. It derives a same-host WebSocket endpoint, accepts a validated `ws://wss:` query override, and exposes connected, reconnecting, offline, and error state. This is configurable local demonstration behavior, not proof of authenticated or secure remote deployment.

6.6 Automated verification

The final project run reports 145 tests passing, covering AHU thermal/electrical semantics, comfort-debt coordination, PID actuator feedback, ecosystem stop/scenarios, strict command validation, retained-command rejection, command-ID replay/conflict handling, optional integrated audit writes/failure visibility, model artifact/OOD/unavailable behavior, exact default-dataset/model/metric reproduction, MQTT payloads, dashboard telemetry/correlation, and presentation helpers. Raw console and JUnit evidence are archived under `release-evidence/project2/`; source state is disclosed as a pre-final-commit working tree and must be refreshed after commit.

A real local classroom-broker run additionally archived correlated accepted stress, pause, and baseline results; constrained grants; paused zero instantaneous values with unchanged cumulative energy; and fresh-subscriber retained recovery. Remaining gaps include browser-level automation, broker-loss recovery, field calibration and multivariate drift tests, deployed authentication/TLS/ACL tests, externally anchored comprehensive audit, and proof that risk changes an autonomous action. The latter absence is expected because model action is intentionally advisory.

Chapter 7

Governance, Ethics, Privacy, and Cybersecurity

7.1 Governance status and formal ethical disposition

This chapter applies the course ethics and cybersecurity frameworks to the complete chain

data → model/policy → prediction or allocation → human decision → simulated action. (7.1)

The present disposition is **PILOT with RESTRICTIONS**: approve the system only as a synthetic classroom pilot and, after the evidence-hardening gate, as a read-only digital shadow. Restrict predictive risk to human decision support. Reject physical autonomy, automatic work-order creation, individual monitoring, and disciplinary or performance uses until the relevant safety, privacy, security, model, fairness, and accountability gates in [table 8.3](#) pass. This is a governance decision, not a claim that planned controls already exist.

Table 7.1: Six-question Digital Twin ethics canvas and current disposition.

Question	Current answer	Required control or decision
Purpose	Explain shared-HVAC contention, comfort, energy, equipment condition, and synthetic maintenance risk.	Limit use to teaching, simulation, evidence development, and a governed read-only pilot.
Data	Simulated aggregate occupancy, room state, AHU/fan state, commands, decisions, and model artifacts. No names, video, faces, or biometrics.	Maintain a field-level inventory, lineage, minimum retention, access rules, and explicit prohibition of secondary surveillance use before real data.
People	Occupants may gain comfort; facilities and maintenance staff may gain visibility. Lower-priority rooms may bear denied airflow; operators may bear false-alert workload; the institution bears cost/security risk.	Measure benefit and burden by room and operating regime; consult affected occupants and operators; define complaint, correction, and escalation routes.
Power	The coordinator allocates simulated flow automatically. Operators control room requests. The model only recommends review. No authenticated global pause or durable appeal workflow exists.	Preserve human authority for maintenance and high-impact control; implement authenticated pause, override, acknowledgement, rationale, and audit before physical use.
Failure	Risks include unfair starvation, stale or spoofed data, false negatives, automation bias, privacy inference, broker compromise, and failed fallback.	Define stop rules, safe degraded mode, incident ownership, evidence preservation, verified recovery, and re-approval after material change.
Decision	Current evidence supports a restricted simulated pilot, not autonomous deployment.	Advance sequentially: evidence hardening → read-only shadow → human-reviewed recommendations → bounded automation only after signed gates.

7.2 Ethical principles, stakeholders, and benefit–burden analysis

The governance decision is evaluated through beneficence, non-maleficence, autonomy, and justice. Beneficence is the potential improvement in comfort visibility, energy understanding, and maintenance lead time. Non-maleficence requires preventing unsafe commands, privacy leakage, false confidence, recurrent room starvation, and over-reliance on synthetic predictions. Autonomy requires intelligible explanations, meaningful human authority, correction and appeal routes, and the ability to stop the system. Justice requires measuring who receives airflow, comfort, attention, and cost savings and who carries monitoring, discomfort, false-alert, or false-negative burdens.

Table 7.2: Stakeholder voice, expected benefit, potential burden, and governance response.

Stakeholder	Potential benefit	Potential burden or right at risk	Required voice/control
Room occupants/students	Improved comfort and transparent room state	Recurrent low allocation; inference of room-use patterns; lack of notice	Plain-language notice, feedback/complaint route, comfort-disparity review, no identity linkage
Facilities operators	Better situational awareness and explainable allocation	Alert fatigue, automation bias, responsibility without authority	Training, alternatives, approval/override authority, rationale capture, escalation procedure
Maintenance staff	Earlier inspection support	False alerts, missed failures, workload displacement	Prospective validation, workload threshold, ability to defer/challenge, outcome feedback
Data/model owners	Reproducible evidence and model improvement	Accountability for drift, weak target semantics, or biased data	Model card, lineage, version approval, monitoring, rollback and re-approval duties
IT/OT security	Defined integration boundary	Expanded attack surface and incident workload	Identity, segmentation, audit, stop authority, response/recovery rehearsal
Institution / building owner	Potential energy/resilience value	Capital cost, liability, privacy and reputation risk	Risk acceptance, benefits review, independent gate approval, transparent communication
Vendors / integrators	Clear interfaces and responsibilities	Supply-chain and privileged-access risk	Time-bounded least privilege, contractual duties, revocation, audit and incident cooperation

No favorable aggregate KPI may conceal a severe burden on one room or stakeholder group. Comfort and denied-flow evidence must therefore be reported alongside energy and maintenance value.

7.3 Purpose limitation, autonomy, and transparency

Permitted current uses are simulation, teaching, explanation of shared-resource coordination, and synthetic predictive-model evaluation. Prohibited uses are individual identification, attendance or productivity scoring, disciplinary action, commercial profiling, automatic maintenance shutdown, and control of real equipment. Linking aggregate occupancy to identities, schedules, access records, or cameras is a material purpose change requiring a new privacy and ethics review.

Transparency exists at two decision layers. The coordinator exposes request, grant, priority components, and reason codes. The risk model exposes version, input telemetry, thresholds,

and signed standardized log-odds terms. Both are inspectable and deterministic given state. Transparency does not establish correctness or causality; it enables review. Audience-specific communication is required:

- occupants receive a plain-language statement of purpose, collected/inferred data, retention, prohibited uses, and complaint/correction route;
- operators receive recommendation, confidence/status, contributing factors, alternatives, freshness, and override/escalation controls;
- auditors receive data lineage, versions, validation, approvals, incidents, and decision/action traces;
- management receives benefits, burdens, residual risks, accountable owners, and explicit go/no-go evidence.

Maintenance recommendations are simulation-only. A human decides whether to inspect equipment, and no work order, setpoint, fan-speed, or airflow change is generated from risk. This is an appropriate risk boundary until real data, safety analysis, security, calibration, OOD handling, and override procedures exist. A production governance process should use a model card and life-cycle controls consistent with risk-management principles such as NIST AI RMF [8, 9].

7.4 Data governance, privacy, consent, and retention

Table 7.3: Prototype data inventory and required production governance.

Data	Purpose/current form	Sensitivity	Production requirement
Aggregate occupancy	control priority; integer 0–30	can reveal room-use patterns despite no identity	purpose limitation, minimum retention, role access, aggregation review
Room environment	comfort/control simulation	operational building data	retention and integrity policy
AHU/fan telemetry	energy, condition, risk	asset/maintenance information	asset owner, quality checks, lineage
Commands and decisions	operation and explanation	security/safety relevant	authenticated identity, tamper-evident audit, retention
Model artifacts	inference and reproducibility	integrity-sensitive IP/configuration	hashes/signatures, approval, rollback

The prototype uses no real data subjects: occupancy and all other operational data are simulated, so individual consent is not applicable at Stage 0. This does not pre-authorize real sensing. Before Stage 2, the data steward must document the lawful/institutional basis, occupant notice and consultation, whether consent or an alternative basis applies, correction/complaint handling, each field’s retention duration, deletion method, sharing recipients, and a prohibition on incompatible secondary use. A new data source, linked identifier, downstream use, or retention extension triggers re-approval.

Aggregate counts reduce data collection but are not automatically anonymous in all contexts; re-identification and inference risk depends on auxiliary information. Production anonymisation and access design should follow local guidance such as Singapore PDPC’s anonymisation guide [11]. Data minimisation should prefer the lowest detail and frequency that supports the approved purpose, bounded histories instead of indefinite retention, and edge aggregation where practical.

7.5 Bias, allocation fairness, and mitigation

Model bias and coordinator fairness are distinct. Predictive subgroup evaluation should compare false-negative, false-positive, calibration, and alert rates by asset, room, season, load, occupancy regime, and equipment age, provided each group has sufficient governed data. The current synthetic rows have no asset or subgroup identity, so no such results can be claimed.

The coordinator can create disparate comfort outcomes through occupancy-first ranking and stable room-ID ties. Proposed operational metrics are denied-flow duration, comfort degree-hours, maximum consecutive starvation, and room-to-room disparity. Before constrained physical automation, governance should define (i) minimum sample sizes; (ii) alert thresholds for model disparity and comfort debt; (iii) an accountable reviewer; (iv) investigation/recalibration/retraining or policy-remediation steps; (v) approval and rollback; and (vi) manual fallback. These are target controls, not implemented evidence.

7.6 Accountability and decision rights

Table 7.4: Minimum decision-rights matrix. Named people must replace role categories in each gate dossier.

Decision	Authority and consultation	Required record
Run/stop classroom simulation	Demo operator; project lead accountable	timestamp, reason, current version
Accept model for advisory pilot	Model owner responsible; maintenance/facilities and independent reviewer consulted; project sponsor accountable	signed model card, validation, threshold/OOD policy
Change allocation or safety policy	Controls owner responsible; facilities/safety and affected operators consulted	versioned policy, hazard/fairness evidence, rollback result
Approve real telemetry	Data steward and facilities owner; privacy/security consulted	data inventory, purpose/basis, retention/access approval
Enable physical automation	Facilities authority accountable; controls safety and cybersecurity must approve independently	commissioned safety envelope, penetration/fallback evidence, residual-risk acceptance
Declare incident/pause system	On-duty operator or cybersecurity owner may pause immediately	incident ID, containment, evidence, notifications
Resume after incident	Facilities and cybersecurity jointly approve; model owner added if model/data affected	clean-state verification, observe-only result, approval

Accountability follows a RACI-like rule: the responsible role operates the control, one accountable authority owns the outcome, affected domain experts are consulted, and impacted people are informed in usable language. Missing names or signatures mean the gate has not passed.

7.7 Cybersecurity lifecycle: Govern–Identify–Protect–Detect–Respond–Recover

Table 7.5: Security lifecycle mapped to current evidence and required gate evidence.

Function	Current evidence	Required before real deployment
Govern	Restrictions and future owner categories are documented.	Named risk owner, appetite, decision rights, vendor/AI/data rules, emergency authority, stop rule, communication owner, signed residual-risk decision.
Identify	Architecture, MQTT contract, data inventory, model artifact, and threat paths are documented.	Owned asset/dependency/vendor inventory, trust boundaries, data classification, exposure/SBOM/secrets inventory, criticality and residual-risk register.
Protect	Topic/type/range checks, command queue, finite capacity, retained state, and LWT provide limited reliability safeguards.	Authenticated identities, TLS/WSS, least-privilege ACLs, segmentation, replay/expiry/rate controls, bounded queues, signed artifacts, minimum-ventilation and local safety constraints.
Detect	Timestamps, online/offline status, freshness labels, and console traces provide demonstration visibility.	Correlated identity/data/model/command/physical logs, integrity and anomaly alerts, model/OOD monitoring, security escalation, tamper-evident retention.
Respond	Operators can stop host processes or use narrow room controls; response duties are documented only.	Independent pause/kill path, incident severity and contact matrix, component isolation, token revocation, evidence preservation, safe manual/degraded mode, communications.
Recover	Simulator can be restarted and retained state helps clients obtain the latest message; this is not recovery assurance.	Clean backups/golden configuration, credential rotation, last-known-good model/policy, phased restore, observe-only validation, independent approval, rehearsal and post-incident update.

7.8 Current security posture and threat register

The default Compose Mosquitto configuration states `allow_anonymous true`, listens on plaintext MQTT port 1883 and WebSockets port 9001, and uses no ACL. A separate `mosquitto-hardened.conf/ACL` example requires TLS certificate and password files, but default Compose does not deploy it; presence is not operating evidence. The simulator strictly rejects oversized/invalid/non-object/non-finite payloads and invalid types/ranges, rejects retained commands, queues non-retained commands for serialized mutation, emits application results, and uses a 1,024-entry process-local fingerprint cache to replay identical IDs without mutation and reject conflicts. Optional `ECOHVAC_AUDIT_PATH` writes sanitized command results to a local hash

chain. These are useful reliability/accountability controls but do not authenticate the self-declared actor or protect transport confidentiality/integrity. Operational-technology security requires defense in depth, segmentation, identity, monitoring, recovery, and safely engineered control behavior [13].

Table 7.6: Threat model: implemented boundary versus production target.

Threat/asset	Example attack path	Impact	Current control/evidence	Required target control and verification
Impersonation	anonymous publish to command topic	unsafe occupancy/set-point/degradation state	schema/type/range validation only	mTLS or per-device credentials; least-privilege ACL; unauthorized publish test
Eavesdrop/tamper	plaintext MQTT and WebSockets (ws://)	privacy, false telemetry/commands	none in transport	TLS/WSS, certificate validation/rotation, cipher and MITM test
Replay	resend valid command	repeated unsafe state transition	no nonce/sequence	timestamp/nonce/sequence, expiry, idempotency, replay test
Retained-message abuse	stale malicious state	misleading clients	commands intended non-retained; telemetry timestamp	broker ACL/policy, freshness enforcement, retained-state reset test
DoS/flooding	high-rate publishes or broker loss	delayed/lost control evidence	command queue and LWT; no bounded rate policy	rate limits, bounded queues, watchdog, safe local fallback, recovery exercise
Topic escalation	wildcard subscribe/publish	confidentiality/control compromise	no ACL	identity-bound allowlists and ACL regression tests
Model substitution	alter JSON coefficients	arbitrary risk/advice	feature schema/array-length checks	SHA/signature, finite/range validation, approved registry, last-known-good rollback
Endpoint exposure	hard-coded localhost/ ws://	failed/unsafe remote deployment	local-only assumption	environment configuration, WSS, origin policy, segmentation
Audit repudiation	actor denies command	weak accountability	optional local hash-chained sanitized command-result record; self-declared actor; no external anchor	authenticated actor, wider decision/action coverage, externally anchored/signature-protected
Supply/deployment drift	mutable image tag/partial Compose	unreviewed change or missing service	Mosquitto v2 tag; only broker packaged	centralized audit and verification pinned digest/SBOM, signed build, full service health checks

Each target control must enter a control register with lifecycle function, owner, priority, verification artifact, due/review date, status, and accepted residual risk. The table above is a threat analysis, not evidence that target controls are operating.

7.9 Stop rules, incident response, and verified recovery

Production go/no-go requires explicit minimum ventilation and equipment limits, stale-data rules, local fallback independent of the broker, manual pause/override, acknowledged commands, and tested rollback. The current prototype has no authenticated global pause or verified recovery workflow. MQTT retained messages and LWT provide state/liveness visibility; they do not provide safe fallback, incident containment, or clean restoration [10].

The minimum incident sequence is:

1. **Triage safety:** determine physical, comfort, privacy, data, model, and service impact.
2. **Contain:** disable automation or affected commands, isolate accounts/tools/network paths, revoke access, and preserve traces.
3. **Fallback:** continue only in the approved manual/read-only/last-known-safe mode.
4. **Investigate and communicate:** record knowns, unknowns, actor/data/model/command lineage, affected stakeholders, and notification duties.
5. **Restore progressively:** rebuild from verified configuration, rotate credentials, validate data/model/policy, and run observe-only before authority is returned.
6. **Learn and re-approve:** update controls, model card, threat register, training, vendor duties, and the affected roadmap gate.

Material changes to data, model, policy, operational context, automation authority, or downstream use invalidate the previous approval and require re-review. A 48-hour manual-operation exercise, recovery-time/recovery-point objectives, and independent restoration evidence are Stage 4/5 requirements, not current achievements.

Chapter 8

Illustrative Business Case and Strategic Roadmap

8.1 ROI formulation

Let annual HVAC energy be E_a (kWh/year), tariff τ (S\$/kWh), assumed reduction r_E , avoided-incident value V_I (S\$/year), annual support cost C_S , and initial implementation cost C_0 . The dashboard implements

$$C_{base} = \max(0, E_a) \max(0, \tau), \quad (8.1)$$

$$S_E = C_{base} \text{ clip}(r_E; 0, 1), \quad (8.2)$$

$$B_{annual} = S_E + \max(0, V_I), \quad (8.3)$$

$$B_{net} = B_{annual} - \max(0, C_S), \quad (8.4)$$

$$\text{ROI}_{1y} = 100(B_{net} - C_0)/C_0, \quad C_0 > 0, \quad (8.5)$$

$$M_{payback} = 12C_0/B_{net}, \quad B_{net} > 0. \quad (8.6)$$

This is first-year simple ROI, not an annualized return. If net benefit is non-positive, payback is economically undefined; the current function returns zero, which should be rendered as “no payback” rather than zero months. The implementation does not separately model labor and downtime; both must be included, if justified, within aggregate avoided-incident value.

8.2 Current illustrative base case

The UI defaults are $E_a = 45,000$ kWh/year, $\tau = \text{S}\$0.30/\text{kWh}$, $r_E = 8\%$, $V_I = \text{S}\$12,000/\text{year}$, $C_S = \text{S}\$4,000/\text{year}$, and $C_0 = \text{S}\$25,000$. They yield $C_{base} = \text{S}\$13,500$, $S_E = \text{S}\$1,080$, $B_{net} = \text{S}\$9,080$, first-year ROI -63.68% , and payback 33.04 months. A negative first-year ROI is not a computation error; it states that one year of modeled benefit does not recover capex.

No facility baseline, tariff document, incident frequency, downtime valuation, quote, or measured reduction is attached. Every value is therefore an editable scenario, not evidence of savings.

8.3 Illustrative sensitivity

Table 8.1: Illustrative low/base/high arithmetic. Inputs are scenarios only and are not forecasts or measured outcomes. “None” means annual net benefit is non-positive.

Input/output	Low	Base	High
Annual energy (kWh)	40,000	45,000	50,000
Tariff (S\$/kWh)	0.25	0.30	0.35
Reduction	4%	8%	15%
Avoided incidents (S\$/yr)	4,000	12,000	30,000
Support (S\$/yr)	6,000	4,000	3,000
Initial cost (S\$)	35,000	25,000	20,000
Energy saving (S\$/yr)	400	1,080	2,625
Net benefit (S\$/yr)	−1,600	9,080	29,625
First-year ROI	−104.57%	−63.68%	48.13%
Payback (months)	None	33.04	8.10

The broad spread shows that avoided incidents dominate the business case. A pilot must replace assumptions with metered baseline energy normalized for weather/occupancy, maintenance records, incident probability/severity, quotes, and support effort. A multi-year decision should add discount rate, cash-flow timing, replacement costs, residual value, and NPV; none is fabricated here.

8.4 Benefits-realization KPIs

A real pilot should pre-register: baseline and post-intervention HVAC kWh; degree-hours outside comfort target; alert precision/recall and lead time; unplanned fan incidents and downtime; operator response time; avoided maintenance cost; telemetry completeness/freshness; and security/fairness exceptions. Energy claims require an agreed measurement and verification method rather than comparing raw simulation totals.

8.5 How to read the deployment roadmap

The roadmap is a sequence of evidence gates, not a claim that deployment has occurred. Only Stage 0 is **Current**. Stages 1–5 are **Planned**. All stated durations are planning estimates that begin only after prerequisites are met; they are neither verified elapsed times nor commitments. Advancement is sequential: a later stage depends on acceptance of every earlier gate. A missing security, data-quality, model, safety, or comfort artifact is a *stop condition*, not an item to defer after launch.

The optional five-room activity appears only in Stage 1 as a proposed software scalability and allocation-fairness experiment. It requires generalized configuration and tests and does not imply that five rooms, five physical zones, or an N -room interface exist now. Production scale in Stage 5 means multiple governed AHUs/sites after earlier evidence is accepted; it is not a continuation of the current two-room screenshot.

8.6 Roadmap execution, change management, and gate authority

A roadmap becomes executable only when role categories are replaced by named people, planning estimates by approved windows, and business assumptions by sourced evidence. Before a planned stage starts, its sponsor must approve a stage charter containing scope, exclusions, dependencies,

named responsible/accountable/consulted/informed roles, calendar dates, an approved cost ceiling or quote range, benefits hypothesis, evidence owners, stop rules, and rollback resources. This report deliberately does not invent names, dates, or costs that the team or facility has not supplied.

Table 8.2: Mandatory execution evidence and authority for each roadmap transition.

Transition	Cost and value evidence	People/change evidence	Go/no-go authority
0 → 1	Effort estimate for submission hardening; no savings claim	Named deliverable owners, current-system pitch rehearsal, defect ownership	Project/team lead accepts complete reproducibility and submission dossier
1 → 2	Approved sensor/integration/security quote range and measurement plan	Facilities, IT/OT, data steward, and occupant representative consulted; support/training plan drafted	Facilities sponsor, data steward, and cybersecurity owner jointly approve read-only access
2 → 3	Pilot budget, operator/maintenance effort, alert-workload ceiling, and evaluation cost	Maintenance workflow mapping, operator training/competency check, feedback and appeal route	Maintenance authority and independent model-risk reviewer approve advisory pilot
3 → 4	Commissioning, safety, security, support, and rollback cost; approved multi-year NPV case	Simulation/tabletop drills, competency sign-off, affected-occupant communication, staffed fallback	Facilities authority approves only after independent controls-safety and cybersecurity approvals
4 → 5	Per-site total-cost and realized-benefit evidence with tolerance/stop threshold	Site consultation, local training, service ownership, vendor responsibilities, incident exercises	Enterprise sponsor plus each site authority; one site cannot approve another

Every gate dossier must record decision date, decision (approve, pilot, redesign, restrict, or reject), approvers, evidence versions/hashes, open exceptions, accepted residual risk, expiry/review date, and rollback owner. Absence of a required signature is a failed gate. A material change to data, model, allocation policy, automation authority, facility, threat environment, or use purpose invalidates the affected approval.

Change management is not limited to a software release. Before Stage 2, stakeholders must understand purpose, collected/inferred data, prohibited uses, and complaint routes. Before Stage 3, maintainers must demonstrate how to interpret, defer, challenge, and record an alert without treating probability as certainty. Before Stage 4, operators must demonstrate manual override, safe degraded operation, incident escalation, and recovery in drills. Training completion, competency assessment, consultation feedback, unresolved concerns, and communications are gate evidence. Post-launch reviews compare actual comfort, workload, incidents, costs, and benefits with pre-registered limits and may trigger redesign or rollback.

Figure 8.1: One-page gated deployment timeline. Stage 0 is verified current scope; every later stage is a proposed estimate and cannot begin until the preceding decision gate passes.

Stage 0		Stage 1		Stage 2		Stage 3		Stage 4		Stage 5
Current	→	Planned	→	Planned	→	Planned	→	Planned	→	Planned
Two-room simulation		Evidence hardening and optional five-room test		Read-only digital shadow		Human-in-the-loop maintenance		Constrained automation		Multi-AHU/site scale
Verified source state; no duration estimate		2–4 weeks estimated		8–12 weeks estimated		8–12 weeks estimated		3–6 months estimated		Phased; estimate set after Stage 4
Evidence: active equations, 145 tests, synthetic model, local demo		Gate 1: deliverables, reproducible tests, security plan, fairness results		Gate 2: calibrated sensors, governed data, baseline, read-only security		Gate 3: prospective model utility, calibration, operator acceptance		Gate 4: safety envelope, fallback, override, penetration and comfort evidence		Gate 5: site-local resilience, load/fairness evidence, approved business case

No gate pass means no advancement. Roll back to the last accepted stage whenever its safety, security, data, model, comfort, or value conditions cease to hold.

8.7 Stage-by-stage plan and decision gates

Table 8.3: Detailed gated plan. Durations and future thresholds are proposed planning assumptions requiring owner approval; only Stage 0 facts are as-built evidence.

Stage/status	Planning field	Definition, evidence, and decision rule
0. Two-room simulated pilot Current	Objective and scope	Demonstrate causal contention between exactly room1 and room2 , deterministic shared-AHU allocation, active supply-air physics, runtime advisory fan risk, and operator-visible evidence. No real facility or five-room implementation is in scope.
	Duration and dependency	As-built source state audited 14–15 August 2026; no elapsed delivery duration is asserted. This is the dependency for every future stage.
	Activities/deliverables and accountable role	Current source, JSON model artifact/expanded metrics, executed notebook, two-room MQTT views, report, figures, tests, generated pitch, and archived real-broker evidence. Accountable role category: teaching/demo owner; named person remains a team assignment. Submission closure still requires completed identity/course metadata, final commit-linked source-state/hash refresh, rehearsal, and a fresh current browser capture tied to the archived transcript.
	KPIs and gate evidence	Verified current facts: 145 tests pass; allocator satisfies $0 \leq g_i \leq r_i$ and $\sum g_i \leq F^{avail}$ in tested cases; stored holdout counts are 92 TP, 31 FP, 63 FN, 294 TN at cutoff 0.5. The notebook executes 11/11 code cells, the pitch validates as eight slides/pages, and the scripted real-broker scenario passes. Gate 0 remains not formally approved : metadata, named authority/reviewer disposition, final commit-linked evidence refresh, and zero unresolved critical demo defects are still required.
	Stop/rollback, ROI, and HW2	If the model, broker, or UI is unavailable, use documented individual commands or stop the demo; do not imply live confirmation. ROI remains editable illustrative arithmetic only. R9 notebook and R10 pitch artifacts are present, while official review and submission remain external actions.
1. Evidence hardening and scalability experiment Planned	Objective and scope	Make the simulated evidence reproducible and submission-complete. Optionally generalize configuration for a <i>proposed five-room software experiment</i> that tests fairness, latency, and message load; five rooms remain future work until code, tests, and evidence exist.
	Estimated duration and dependency	2–4 weeks estimated after Gate 0 artifacts are complete. Depends on the accepted two-room baseline and a frozen scenario/model version. Calendar estimate is unverified.

Stage/status	Planning field	Definition, evidence, and decision rule
	Activities/deliverables and accountable role	Exercise the implemented guided scenarios end to end; verify the executed notebook in a pinned environment; add commit/build record, JUnit/coverage, scenario manifests, telemetry logs, browser/broker integration tests, multivariate drift/OOD tests, and fairness metrics. Produce the final current-system Project 2 executive pitch as both PPTX and exported PDF. The team must name a pitch owner, audience (instructor plus executive/facilities decision makers), and due date no later than the Gate 1 submission-readiness review. Optional five-room work requires configuration-driven room creation rather than copied UI cards. Accountable role categories: simulation lead, ML owner, test/evidence owner, cybersecurity reviewer, and named pitch owner.
	Pitch acceptance evidence	The PPTX/PDF must open correctly, use only current Project 2 screenshots, fit the assigned presentation time, and cover: problem/business need; two-room architecture and centralized coordination; live contention evidence; predictive model and honest synthetic/false-negative/temporal limitations; ROI assumptions and sensitivity; current-versus-target security; ethics, privacy, bias and human oversight; gated roadmap; and a specific pilot ask. Evidence is the final PPTX/PDF, rehearsal against the time limit, content checklist, and team/instructor-readiness sign-off. The Markdown outline, Project 1 PPTX, and legacy videos do not pass this gate.
	KPIs/thresholds, prerequisites, and gate evidence	Proposed Gate 1 thresholds: 100% pass of the frozen automated suite; reproducible scenario outputs from the same seed/manifest; no unresolved critical/high demo-security finding; all R9/R10 files open and execute; every model metric regenerated from code; allocator capacity invariants pass for 2 and, only if implemented, 5 rooms; report maximum starvation duration and allocation disparity rather than claiming fairness. Required evidence: logs, hashes, notebook outputs, final pitch PPTX/PDF and checklist, defect register, and reviewer sign-off. No final pitch or notebook means no submission-readiness pass.
	Stop/rollback, ROI, and HW2	Stop expansion if generalized code changes two-room behavior, fairness/starvation is unmeasured, or guided controls remain unverifiable; revert to the frozen two-room baseline. ROI maturity: replace UI arithmetic with a documented measurement plan and source list, not new savings claims. Closes R9/R10 and strengthens R1–R8 evidence; optional five-room work tests R3/R4/R7 but is not required to claim the current system.
2. Read-only facility digital shadow Planned	Objective and scope	Connect governed real telemetry without issuing building commands, establish whether simulated variables map to actual assets, and measure data quality and baseline comfort/energy. Scope is one approved AHU and its mapped zones; control remains read-only.
	Estimated duration and dependency	8–12 weeks estimated after Gate 1. Depends on approved data ownership, asset/topic mapping, security architecture, sensor access, and the frozen shadow software. Calendar estimate is unverified.

Stage/status	Planning field	Definition, evidence, and decision rule
	Activities/deliverables and accountable role	Calibrate/map sensors; document lineage, retention, access and consent/purpose; establish authenticated encrypted ingestion and segmentation; record weather/occupancy-normalized energy and comfort baselines; execute missing/stale/OOD tests; create incident and disconnect procedures. Accountable role categories: facilities owner, data steward, cybersecurity owner, and measurement-and-verification lead.
	KPIs/thresholds, prerequisites, and gate evidence	Proposed Gate 2 thresholds: $\geq 95\%$ telemetry completeness over the agreed window; timestamp/asset mapping verified; calibration error limits specified and met per sensor class; zero unauthorized command route from shadow to plant; all critical/high security findings closed; baseline duration and normalization method approved; missing/stale data visibly degraded rather than treated as live. Evidence: calibration records, access/ACL tests, data-quality report, threat-model test results, baseline dataset, and facilities/data/cyber sign-off.
	Stop/rollback, ROI, and HW2	Disconnect ingestion on privacy breach, command-path discovery, persistent quality below threshold, unsafe load on facility systems, or unresolved critical/high vulnerability; revert to offline simulation. ROI maturity: replace annual-energy/tariff assumptions with governed baseline and tariff evidence while incident avoidance remains unclaimed. Advances R5/R6/R8 and provides external evidence needed to revisit R2/R3.
3. Human-in-the-loop maintenance pilot Planned	Objective and scope	Evaluate advisory alerts prospectively while qualified staff retain all maintenance authority. The model may recommend review; it may not create an autonomous work order or alter airflow, setpoint, or fan operation.
	Estimated duration and dependency	8–12 weeks estimated after Gate 2, or longer if too few relevant events occur. Depends on accepted data quality, a governed target definition, model card, frozen threshold policy, operator protocol, and ethics/privacy approval. Calendar estimate is unverified.
	Activities/deliverables and accountable role	Train/recalibrate with asset/time-separated data; compare baselines; assess calibration, PR/ROC, confidence intervals, lead time, subgroup/OOD performance and alert workload; run prospective shadow alerts; record operator acceptance, deferral, reason, and outcome; test rollback. Accountable role categories: model owner, facilities maintenance owner, data steward, and model-risk reviewer.
	KPIs/thresholds, prerequisites, and gate evidence	Before the pilot, owners must set numerical minimum recall/precision or cost-based thresholds, calibration tolerance, maximum false-alert workload, minimum sample/event counts, subgroup disparity limits, and OOD/staleness behavior; this report does not invent those site-specific values. Gate 3 requires all pre-registered thresholds to pass with uncertainty intervals, no unresolved critical subgroup/safety issue, demonstrated operator utility, complete audit records, and independent model/operations approval. Evidence includes a prospective evaluation report and signed model card.

Stage/status	Planning field	Definition, evidence, and decision rule
	Stop/rollback, ROI, and HW2	Suppress alerts and revert to conventional monitoring if calibration/recall falls outside approved limits, drift/OOD alarms persist, operator workload exceeds its threshold, unsafe advice occurs, or auditability fails. ROI maturity: estimate alert-handling labor and observed maintenance outcomes, but claim avoided incidents only with an approved counterfactual method. Strengthens R2/R7/R8; R1 control remains deterministic and separate.
4. Constrained physical automation Planned	Objective and scope	Permit only bounded, reversible actions inside an independently engineered safety envelope. This is not approval for unrestricted autonomous optimization or model-directed maintenance.
	Estimated duration and dependency	3–6 months estimated after Gate 3. Depends on accepted prospective model evidence, controls hazard analysis, minimum-ventilation requirements, authenticated command architecture, manual override, fail-safe local control, change approval, and rollback rehearsal. Calendar estimate is unverified.
	Activities/deliverables and accountable role	Implement bounded command allowlists, acknowledgements, freshness/sequence/replay controls, mTLS/credentials/ACLs, local fallback/watchdogs, hard limits, comfort debt/fairness safeguards, immutable audit, staged commissioning, penetration testing, and operator training. Accountable role categories: controls/safety owner, facilities authority, cybersecurity owner, and independent commissioning reviewer.
	KPIs/thresholds, prerequisites, and gate evidence	Proposed Gate 4 minimums: zero unresolved critical/high safety or security finding; 100% tested manual-override and rollback success in commissioned cases; stale/broker/OOD tests always enter the approved fallback; no minimum-ventilation violation; comfort and starvation limits remain within owner-approved thresholds; every command is authenticated, acknowledged, and auditable. Evidence: hazard analysis, commissioning results, penetration report, fallback drills, comfort/ventilation report, and formal facilities/safety/cyber go/no-go.
	Stop/rollback, ROI, and HW2	Hard-disable automation on limit violation, lost acknowledgement, stale/OOD input, broker/security incident, comfort threshold breach, unexplained model drift, or failed audit; return to local conventional control and last-known-good software. ROI maturity: measure incremental energy, labor, downtime, commissioning/support cost and uncertainty; require an approved multi-year NPV/business case before wider rollout. Advances R1/R5/R6/R7/R8 under real constraints.
5. Multi-AHU/site production scale Planned	Objective and scope	Scale only the accepted bounded design across separately approvable AHUs/sites with local resilience and federated governance. This is production-target planning, not evidence of current scale and not the optional five-room simulation experiment.
	Estimated duration and dependency	Phased estimate to be set only after Gate 4 evidence and site discovery; no defensible calendar is asserted now. Each site depends on its own mapping, baseline, hazards, security review, and local go/no-go.

Stage/status	Planning field	Definition, evidence, and decision rule
	Activities/deliverables and accountable role	Package pinned/signed services and SBOM; validate tenant/site isolation, capacity/load, failover, disaster recovery, observability, model registry/rollback, local policy ownership, cross-site fairness reporting, support and incident response. Accountable role categories: platform owner, site facilities authority, enterprise cybersecurity, model governance, finance, and service owner.
	KPIs/thresholds, prerequisites, and gate evidence	Gate 5 thresholds must be contracted per site: approved uptime/recovery objectives, telemetry completeness, comfort/ventilation limits, incident and vulnerability limits, model and subgroup performance, allocation fairness, support response, and realized-value thresholds. Advancement is site-by-site only after load/failover/security tests, local shadow/HITL evidence, trained operators, signed service ownership, and approved business case. Passing one site never waives another site's gate.
	Stop/rollback, ROI, and HW2	Isolate or revert an affected site on safety/security/data/model/comfort breach; prevent cascade by retaining local operation. Pause further rollout if realized benefits do not meet approved stage thresholds or total cost exceeds the approved tolerance. ROI maturity: compare verified cash flows with the approved NPV case and revise/stop rather than preserving sunk-cost assumptions. Completes the target intent of R5–R8 only when evidenced; it does not retroactively strengthen the current HW2 claim.

8.8 Cross-stage dependency and gate rule

The dependency chain is

$$\begin{aligned} &\text{two-room evidence} \rightarrow \text{hardened reproducibility} \rightarrow \text{governed shadow data} \\ &\rightarrow \text{prospective human review} \rightarrow \text{bounded automation} \rightarrow \text{site-by-site scale} \end{aligned} \quad (8.7)$$

Each arrow is conditional. A gate dossier must contain the required technical artifacts, KPI results, open-defect disposition, ethics and benefit–burden review, stakeholder consultation and competency evidence, approved cost range, named approving authorities, and rollback test. Decision records use the dispositions *approve*, *pilot*, *redesign*, *restrict*, or *reject*; approval without named signatures and evidence versions is invalid. No composite score may compensate for a missing mandatory domain: for example, favorable ROI cannot offset failed minimum ventilation; high accuracy cannot offset poor calibration or unacceptable false negatives; and encrypted transport cannot offset absent command authorization. If evidence expires because the software, model, facility, threat environment, use purpose, or operating policy changes materially, the affected gate must be repeated.

Chapter 9

Discussion, Threats to Validity, and Conclusion

9.1 What the evidence establishes

The source and captured state establish a real multi-room interaction: local PIDs form separate demands, one finite AHU capacity is calculated from shared condition, a deterministic service allocates less than total demand, and granted flow changes next room state. The architecture is centrally coordinated with local control. The system is autonomous at the deterministic control-simulation level because this loop repeats without a human choosing each grant. A versioned trained logistic model is loaded and scored at runtime, and its probability, band, version, inputs, and contributions are exposed. This is stronger than a threshold-only alert and stronger than independent room visualizations.

The model is, however, predictive decision support. Since scoring occurs after allocation and its output has no command path to the coordinator, it cannot be credited with an autonomous protective action. The strongest accurate claim is:

A working, explainable, centrally coordinated two-room digital-twin simulation with autonomous deterministic control coordination, runtime predictive-maintenance decision support, and a governed roadmap toward constrained production autonomy.

9.2 Internal validity

Equations are traced to active code, and pure coordinator/model components support deterministic tests. Yet a screenshot does not record all initial conditions, run duration, random occupancy path, or repeated outcomes. Guided scenario and correlated application-result controls are now implemented, but the historical capture predates that path and cannot prove it. Runtime stores/publishes the canonical pre-state thermal cooling used by the tick, while PID receives applied-actuator feedback after allocation. Expanded metrics and notebook outputs are reproducible synthetic evidence only. Future experiments still require scenario manifests, raw telemetry, and browser/real-broker traces.

9.3 Construct validity

The thermal/humidity/degradation models use convenient constants without calibration. “Failure within seven days” is a snapshot Bernoulli label, not a temporal event. “Health” is a formula rather than a measured construct. Runtime bands are presentation thresholds, not cost-optimized decisions. ROI inputs are assumptions. These constructs are useful for explaining a mechanism but cannot support field-performance claims.

9.4 External validity

Two simulated rooms and one synthetic fan cannot generalize to different buildings, seasons, ventilation regulations, AHU configurations, equipment makes, or maintenance practices. Anonymous local MQTT and localhost clients do not generalize to production networking. Asset/time data, a read-only digital-shadow stage, subgroup validation, and site-specific controls are required.

9.5 Reliability and reproducibility

The data/model generation seed, equations, parameters, JSON artifact, and test command are inspectable. The source package includes a bibliography, copied current figures, build configuration, and hash manifest. Remaining weaknesses are lack of a recorded clean notebook rerun in the final submission environment, no pinned full dependency lock cited in this report, mutable Mosquitto image tag, host-only services, unrecorded commit/hash in earlier screenshots, and author/course placeholders.

9.6 Ethical and deployment disposition

Applying the evidence and governance analysis yields three explicit decisions:

- **PILOT:** permit the current synthetic two-room simulation for teaching and evidence development; permit a read-only facility shadow only after Stage 1 and Gate 2 prerequisites pass.
- **RESTRICT:** keep fan-risk output advisory to qualified humans, with no automatic work order, equipment command, or employment/disciplinary use. Keep occupancy aggregate and prohibit identity linkage or secondary surveillance.
- **REJECT:** do not enable physical autonomy, unrestricted optimization, model-directed maintenance, or multi-site production until the mandatory safety, security, data, model, fairness, human-oversight, change-management, and recovery gates pass with named approval.

Documentation of a target control is not evidence that the control has been implemented, tested, or assigned. Missing signatures, evidence, or stop/rollback capability mean no advancement.

9.7 Submission-critical disposition

The formal report improves rigor but does not replace named deliverables. Before submission the team should: complete metadata; verify the present executed training notebook reruns cleanly in the submitted environment; create the final current-system Project 2 executive pitch in PPTX and exported PDF from the existing outline and the checklist in [table 8.3](#); capture one scripted end-to-end guided-scenario/acknowledgement/stop run; update stale README counts/architecture wording; and ensure current report/evidence are in the submitted archive. The LaTeX report does not substitute for the named notebook or pitch. Legacy Project 1 PPTX, screenshots, scripts, and videos must not be represented as current Project 2 proof.

9.8 Conclusion

The two-room scope makes more engineering sense than an unsupported five-room expansion: it already proves resource contention and cross-twin causality while leaving time to close higher-value HW2 gaps. The transparent coordinator satisfies capacity invariants, publishes reasons, accumulates bounded comfort debt, and feeds granted actuator output back to each PID, though it still needs explicit minimum-ventilation and stronger fairness guarantees for scale. The logistic model is a real trained runtime artifact with measurable synthetic holdout performance, but its circular generator, weak temporal semantics, false-negative rate, and missing calibration/OOD

treatment constrain its claim. Security and ROI are honestly separated into present prototype state and governed target evidence. The resulting system is a credible classroom ecosystem demonstrator and a defensible basis for a shadow pilot, not a production-ready autonomous facility controller.

Appendix A

MQTT Topic and Command Contract

Telemetry/state topics are retained unless stated otherwise; commands must be non-retained. Payloads are JSON except plain-text status. Values are rounded for transport. MQTT itself does not authenticate these messages in the default classroom broker.

Topic	Direction	Principal fields/purpose
<code>twin/{room}/temperature</code>	sim → client	sensor, value (°C), unit, UTC timestamp
<code>.../humidity, .../occupancy</code>	sim → client	percentage-point display/people with timestamp
<code>twin/{room}/hvac/state</code>	sim → client	on, PID output, setpoint, request, delivery, allocation fraction
<code>twin/{room}/hvac/allocation</code>	sim → client	request, grant, allocation, comfort debt, limited-service time, priority, reasons
<code>twin/{room}/energy</code>	sim → client	stored sensible thermal cooling W used by the tick
<code>twin/{room}/cmd/hvac</code>	client → sim	command=on off ; optional command_id , source
<code>.../cmd/occupancy</code>	client → sim	exact integer value , 0–30
<code>.../cmd/setpoint</code>	client → sim	finite numeric value , 18–30 °C
<code>.../cmd/mode</code>	client → sim	auto or manual
<code>.../cmd/timescale</code>	client → sim	exact integer choice 1, 2, 5, or 10; applied to both rooms
<code>twin/ahu/state</code>	sim → client	maximum/available/delivered flow, supply temperature, clog, speed
<code>twin/ahu/energy</code>	sim → client	thermal cooling; fan/cooling/total electrical W; cumulative electrical kWh; illustrative tariff/cost
<code>twin/ahu/fan/health</code>	sim → client	health, wear, scored risk or explicit abstained/OOD/unavailable status, contributors, telemetry
<code>twin/ahu/coordinator/decision</code>	sim → client	occupied-comfort-debt-v2 , capacity, request, constrained flag, decisions
<code>twin/ahu/cmd/filter_clog</code>	client → sim	finite value 0–1 (physics caps effective clog at 0.95)
<code>twin/ahu/cmd/fan_wear</code>	client → sim	finite value 0–1
<code>twin/ecosystem/cmd/scenario</code>	client → sim	baseline or shared_capacity_stress preset
<code>twin/ecosystem/cmd/simulation</code>	client → sim	pause , resume , or emergency_stop
<code>twin/ecosystem/command/result</code>	sim → client, not retained	command correlation, target, accepted/changed/reason, applied values, duplicate/replay/conflict state where applicable
<code>twin/ecosystem/scenario/state</code>	sim → client	retained scenario identity/revision/phase and operating mode

Topic	Direction	Principal fields/purpose
<code>twin/ecosystem/presentation/state</code>	sim → client	retained coherent snapshot with run/snapshot identity
<code>twin/ecosystem/status</code>	sim → client	retained online/offline; offline configured as LWT

Acknowledgement, replay, and audit behavior

The strict decoder rejects oversized, invalid UTF-8/JSON, non-object and non-finite payloads, invalid identifiers, and unsupported values. The publisher rejects retained commands before queueing. A non-retained command is applied on the simulation thread and emits an application result distinct from MQTT transport PUBACK. Up to 1,024 command IDs are cached in memory: the identical topic/payload is replayed without mutation, while reuse with a different request fingerprint is rejected as `command_id_conflict`. This is bounded process-local idempotency, not durable nonce/expiry protection.

When `ECOHVAC_AUDIT_PATH` is configured, the publisher appends sanitized request metadata (topic, ID/source, retain flag, payload hash and byte count) plus the command result to the local SQLite SHA-256 chain. Audit startup/write failures are visible, including in the command result for write failure. The default stack does not enable this path; the chain is not externally anchored or signed and does not authenticate the self-declared source.

Appendix B

Model Card: fan-risk-logic-v1

Field	Statement
Model	Standardized binary logistic regression, five features, JSON serialized; trained in NumPy by full-batch gradient descent.
Intended use	Teaching demonstration of synthetic fan-condition failure-risk scoring and interpretable operator support.
Out-of-scope use	Real maintenance prediction, safety control, autonomous work orders, regulatory/contractual decisions, or probability claims about real assets.
Training data	2,400 deterministic independent synthetic snapshots, seed 20260805; 1,920 train and 480 random holdout.
Features	filter-clog fraction, fan-speed fraction, vibration mm/s, bearing temperature °C, run hours.
Target	Synthetic Bernoulli label named <code>failure_within_7d</code> ; no seven-day trajectory or observed failure.
Preprocessing	Train-only mean/standard-deviation scaling.
Training	2,400 iterations, learning rate 0.08, no regularization.
Output	$p = \sigma(b + \sum \beta_j z_j)$ clipped at logit ± 50 ; low < 0.35 , medium $[0.35, 0.65)$, high ≥ 0.65 .
Stored evidence	At cutoff 0.5: accuracy 0.8042, balanced accuracy 0.7491, precision 0.7480, recall 0.5935, specificity 0.9046, F1 0.6619, ROC-AUC 0.8449, Brier 0.1467, log loss 0.4471; TP 92, FP 31, FN 63, TN 294.
Explanation	Signed standardized log-odds terms; top three sorted by signed value. Associative, not causal.
Known limitations	Generator circularity; snapshot horizon; one random split; no grouped/temporal/external validation, confidence intervals, field calibration, multivariate drift/OOD policy, cost-based threshold, or subgroup analysis. Finite/missing and univariate domain violations do abstain.
Human oversight	Recommendation only. Risk does not alter coordinator, fan speed, setpoint, airflow limit, or maintenance state.
Required validation	Clean notebook rerun; realistic trajectories/assets; PR analysis; field calibration; interval estimates; threshold cost; multivariate drift/OOD; prospective shadow trial.
Governance	Version, feature schema, means/scales, coefficients and thresholds are inspectable; production requires ownership, approval, signed artifact, monitoring and rollback.

Appendix C

Requirements and Evidence Traceability

ID	Implementation	Evidence	Residual action
R1	Ecosystem tick, PID, and active supply-air updates	chapters 3 and 4 and figure 6.1	scenario manifest and longitudinal metrics
R2	generator, trainer, model JSON, <code>fan_health.py</code> , tick inference	chapters B and 5 and figure 6.2	required notebook and stronger temporal/field validity
R3	two room runtimes sharing AHU and allocator	figures 2.1 and 6.1	repeat contention experiment
R4	architecture docs and <code>occupied-comfort-debt-v2</code>	chapters 2 and 4	keep diagrams/captions synchronized with debt and actuator feedback
R5	command validation/queue and retained state/LWT; anonymous plaintext broker	chapter 7 and tables 7.5 and 7.6	implement/test identity, TLS/WSS, ACL, segmentation, replay/rate, audit and recovery controls
R6	Dashboard ROI function (presentation module)	chapter 8, figure 6.3, and table 8.2	source facility inputs; stage cost ranges; sensitivity/NPV and measurement plan
R7	model terms, thresholds and reason codes; ethics canvas, stakeholder and decision-right analysis	chapters 5 and 7 and tables 7.1, 7.2 and 7.4	implement subgroup/fairness, abstention/OOD, appeal and durable review controls
R8	six-stage Stage 0–5 gated roadmap	tables 8.2 and 8.3	replace role categories/estimates with names, dates, approved cost bands, consultation and signed gate records
R9	scripts, CSV, artifact, expanded metrics, fully executed notebook	notebook EDA/baseline/calibration/OOD and exact reproduction outputs inspected	verify clean rerun and preserve submitted environment/provenance

ID	Implementation	Evidence	Residual action
R10	Executive-pitch Markdown outline only; no final Project 2 deck/export	report figures provide current source evidence only	name pitch owner/audience/deadline; create and rehearse current Project 2 PPTX/PDF to the Stage 1 checklist; exclude legacy Project 1 PPTX/video

C.1 Session 14–16 framework traceability

Table C.2: Course-framework concepts made explicit in the report and their implementation boundary.

Framework element	Report evidence	Current implementation boundary
Data–model–prediction–decision–action chain	chapter 7	Data, allocation and model inference run in simulation; predictive output has no action path
Six-question ethics canvas and four principles	tables 7.1 and 7.2	Governance analysis only; complaint/appeal and disparity safeguards remain planned
Formal deployment disposition	chapters 7 and 9	Restricted synthetic pilot; physical autonomy explicitly rejected pending gates
Decision rights and human authority	table 7.4	Local room commands exist; authenticated pause, durable human review and signed approval do not
Govern–Identify–Protect–Detect–Respond–Recover	tables 7.5 and 7.6	Prototype reliability controls only; production security/response/recovery controls are targets
Evidence-gated strategic roadmap	tables 8.2 and 8.3	Stage 0 current; all later stages planned and dependent on named, signed evidence

Appendix D

Reproducibility and Submission Manifest

D.1 Environment and commands

The Python project declares Python ≥ 3.10 and Streamlit, Paho MQTT, Plotly, NumPy, and pytest development dependencies. Mosquitto is launched through Compose; simulator/dashboard/3D server require separate host commands. The LaTeX build was produced with TeX Live 2025 and latexmk.

Listing D.1: Core reproduction commands (from the Simulation_Room root unless noted).

```
# Broker
docker compose up -d
# Regenerate deterministic data/model/metrics
uv run python simulator/train_fan_model.py --seed 20260805 --rows 2400
# Verify Python suite
uv run pytest -q
# Compile, verify and package report
cd report/academic-report
make distclean
make all
make verify
make package
make verify-package
```

D.2 Known pre-submission placeholders and missing evidence

1. Replace bracketed team/student-ID and course/instructor fields in `metadata.tex`.
2. Record final branch, report commit hash, clean/dirty status, operating system, Python and package versions after the report commit is created.
3. Retain raw test output/JUnit and coverage tied to that final commit if required by the submission channel.
4. Verify the present executed training notebook reruns cleanly in the submitted environment and preserve its outputs/provenance.
5. Add the final current-system Project 2 executive pitch in PPTX and exported PDF; the Markdown outline exists but is not the named deliverable, and legacy Project 1 PPTX/video are invalid evidence.
6. Repair/remove guided scenarios, then capture one command manifest and raw telemetry log.

D.3 Figure provenance

Only five current Project 2 images are copied into this package: integrated architecture, two-room stress, predictive intelligence, strategy/governance, and two-room 3D. The generic operate-dashboard capture and all legacy Project 1 screenshots, deck, scripts, and videos are excluded deliberately.

D.4 Integrity manifest and source archive

The canonical source directory contains the modular LaTeX files, bibliography, five figures, build configuration, final PDF, source ZIP, and SHA256SUMS. The source ZIP contains only the files needed to rebuild the report; it excludes generated LaTeX intermediates, the PDF, the ZIP itself, and the checksum manifest. **make package** recreates the source ZIP and then writes hashes for both the PDF and ZIP plus every source file. **make verify-package** verifies all hashes and tests the ZIP archive. If any source, figure, or metadata changes, rebuild the PDF and rerun both targets before publication.

Appendix E

Parameter and Assumption Register

Parameter	Value	Unit/range	Provenance/status
Room heat capacity C_r	25,000	J/°C	code teaching assumption
Occupant sensible heat q_p	100	W/person	code teaching assumption
Outdoor temperature T_o	32	°C	code fixed boundary
Wall conductance K_w	0.05	W/°C	code teaching assumption; uncalibrated
Supply temperature T_s	16	°C	AHU default
Air density ρ_a	1.2	kg/m ³	code nominal assumption
Air specific heat $c_{p,a}$	1005	J/(kg °C)	code nominal assumption
Room airflow cap	0.16	m ³ /s	ecosystem constant
AHU nameplate flow	0.24	m ³ /s	AHU default
Initial clog/wear	0.05 / 0.03	fractions	AHU defaults
PID K_p, K_i, K_d	0.40, 0.05, 0.05	implementation units	code defaults; no tuning study
Integral bound	±20	error-seconds	clamping only
Temperature bounds	15–40	°C	simulation clipping
Humidity bounds	15–80	percentage points	simulation clipping; proxy model
Occupancy bounds	0–30	people	simulation command/random walk
COP	3.2	ratio	code assumption
Fan reference power	180	W	code assumption
Risk thresholds	0.35 / 0.65	probability	artifact operational bands
Classification threshold	0.5	probability	metric function
Generator rows/seed	2,400 / 20260805	count / integer	deterministic workflow
Learning rate/iterations	0.08 / 2,400	scalar / count	training defaults
Tariff	0.30	\$\$/kWh	illustrative UI assumption
Annual baseline	45,000	kWh/year	illustrative UI assumption
Energy reduction	8	percent	illustrative UI assumption
Avoided incident value	12,000	\$\$/year	illustrative UI assumption
Initial/support cost	25,000 / 4,000	\$\$/ \$\$/year	illustrative UI assumptions

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